# Electroseismic wave properties

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In a porous material saturated by a fluid electrolyte, mechanical and electromagnetic disturbances are coupled. The coupling is due to an excess of electrolyte ions that exist in a fluid layer near the grain surfaces within the material; i.e., the coupling is electrokinetic in nature. The governing equations controlling the coupled electromagnetic-seismic (or ''electroseismic'') wave propagation are presented for a general anisotropic and heterogeneous porous material. Uniqueness is derived as well as the statements of energy conservation and reciprocity. Representation integrals for the various wave fields are derived that require, in general, nine different Green's tensors. For the special case of an isotropic and homogeneous wholespace, both the plane-wave and the point-source responses are obtained. Finally, the boundary conditions that hold at interfaces in the porous material are derived. 

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## INTRODUCTION

Thompson and Gist (1993), Butler *et al.* (1994), and Mikhailov and Haartsen (1996) have all collected field data demonstrating that when seismic waves propagate through the near-surface sediment layers of the Earth (depths less than, say, 300 m), electromagnetic disturbances are generated that can be recorded at the surface. Thompson and Gist (1993) and Pride (1994) have hypothesized that the most likely coupling mechanism is electrokinetic in nature. As is briefly discussed in Sec. I, piezoelectricity cannot be the coupling mechanism.

Starting from first principles, Pride (1994) has derived equations that control such ''electroseismic'' phenomena when an electrokinetic-coupling mechanism is assumed. The basic idea is as follows. The grains within sedimentary materials have an excess charge adsorbed to their surface (usually negative) due to the termination of the crystal lattice and certain chemical reactions that occur between the electrolyte ions and terminal sites on the crystal surface. This chemically bound charge is immobile but is balanced by a layer of free counterions in the adjacent fluid. This layer of fluid containing the mobile counterions is called the ''diffuse-double layer.'' When a seismic wave propagates through the sediments, a relative fluid–solid motion is induced that transports the counterions relative to the fixed charge thus inducing an electrical ''streaming'' current. This streaming current is the source for the electromagnetic coupling.

For compressional waves in a homogeneous porous material, the streaming current leads to a buildup of counter charge in the rarefactions of the wave and a depletion in the compressions. The resultant electric field drives a conduction current that exactly balances the streaming current thus fixing the amplitude of the charge separation within the wave. In this case, as the compressional wave propagates through the medium, there is a constant electric field fixed to the wave with no extent outside of the wave. Independently propagating electromagnetic waves are not generated. However, when the compressional wave is incident at an interface, there is a variation in the streaming-current distribution while the wave is traversing the interface leading to a charge separation that varies with the time signature of the wave. This time-varying charge separation does generate an electromagnetic disturbance that will independently propagate to the Earth's surface where it can be recorded. Somewhat similar comments can be made concerning shear waves (see Sec. II below). These ideas will be elaborated upon and quantified in both this and a forthcoming publication (Haartsen and Pride, 1996).

The overall goal of this paper is to develop the necessary background theory required to solve electroseismic boundary-value problems. The structure of the paper is as follows. In Sec. I, the governing equations for a general anisotropic and heterogeneous porous continuum are presented and discussed. The general statements of energy conservation, uniqueness, and reciprocity are then given as well as the representation integrals for the various wave fields. In Sec. II, we specialize to the case of a homogeneous isotropic whole space. Both the plane-wave and point-source responses are obtained for this special case. Finally, in Sec. III, the boundary conditions that hold at an interface in the porous continuum are derived.

## I. GENERAL PROPERTIES

### A. Governing equations

Assuming an $e^{-i\omega t}$ time dependence, the equations governing the coupled electromagnetic-acoustic wave propagation in a general anisotropic and heterogeneous porous continuum are

$$\nabla \times \mathbf{E} = i\omega \mathbf{B}, \tag{1}$$

$$\nabla\times\mathbf{H}=-i\omega\mathbf{D}+\mathbf{J}+\mathbf{C}, \tag{2}$$

$$\nabla\cdot\boldsymbol{\tau}=-i\omega(\rho\dot{\mathbf{u}}+\rho_f\dot{\mathbf{w}})-\mathbf{F}, \tag{3}$$

$$\mathbf{J}=\mathbf{S}\cdot\mathbf{E}+\mathbf{L}\cdot(-\nabla p+i\omega\rho_f\dot{\mathbf{u}}+\mathbf{f}), \tag{4}$$

$$\dot{\mathbf{w}}=\mathbf{L}\cdot\mathbf{E}+\frac{\mathbf{K}}{\eta}\cdot(-\nabla p+i\omega\rho_f\dot{\mathbf{u}}+\mathbf{f}), \tag{5}$$

$$\boldsymbol{\tau}={}_4\mathscr{S}:\nabla\mathbf{u}+\mathscr{C}\nabla\cdot\mathbf{w}, \tag{6}$$

$$-p=\mathscr{C}:\nabla\mathbf{u}+M\nabla\cdot\mathbf{w}, \tag{7}$$

$$\mathbf{B}=\boldsymbol{\mu}\cdot\mathbf{H}, \tag{8}$$

$$\mathbf{D}=\boldsymbol{\epsilon}\cdot\mathbf{E}. \tag{9}$$

The notation $\dot{\mathbf{u}}(\omega)=-i\omega\mathbf{u}(\omega)$ is being employed. These equations are the natural generalization to the anisotropic case of the isotropic equations obtained by Pride (1994) using volume averaging. The equations naturally group into the four sets following: Eqs. (1)–(3) are the dynamical equations; Eqs. (4)–(5) are the transport equations; Eqs. (6)–(7) are the poroelastic constitutive laws; and Eqs. (8)–(9) are the electromagnetic constitutive laws. These are linear equations so that the wave fields obey superposition. There is a limitation on the maximum frequency that can be treated defined by the condition that wavelengths must be much greater than grain sizes. For sediments in the Earth, this maximum frequency is on the order of $10^6$ Hz.

All coupling is in the transport equations and is due to the electrokinetic-coupling tensor $\mathbf{L}$. If $\mathbf{L}$ were set to zero, these equations would separate into Maxwell's and Biot's (1956) equations. Despite the allowance for anisotropy, there is no allowance for piezoelectric coupling. This is justified when the fluid is an electrolyte because fluid cations will accumulate on the negatively polarized portion of a grain surface and anions on the positive portion thus screening the piezoelectric field of each grain. Such ionic accumulations will take place on time scales of $\epsilon_f/\sigma_f$, where $\epsilon_f$ is the fluid's permittivity ($\sim 10^{-9}$ F/m for water-based electrolytes) and $\sigma_f$ is the fluid's conductivity ($>10^{-3}$ S/m for electrolytes of interest). Thus the screening time will always be $<10^{-6}$ s which is faster than wave periods of interest so that the piezoelectric fields of the grains always remain screened.

If $\mathbf{u}_f(\mathbf{r})$ and $\mathbf{u}_s(\mathbf{r})$ are defined, respectively, as the average displacement of the fluid and solid phases within an averaging volume $V_A$ positioned at point $\mathbf{r}$ ($V_A$ being larger than the grains but much smaller than the wavelengths), then the $\mathbf{u}$ and $\mathbf{w}$ in Eqs. (3)–(7) are defined $\mathbf{u}\equiv\mathbf{u}_s$ and $\mathbf{w}\equiv\phi(\mathbf{u}_f-\mathbf{u}_s)$, where $\phi$ is porosity. The filtration velocity $\dot{\mathbf{w}}(\omega)=-i\omega\mathbf{w}(\omega)$ represents the volume of fluid traversing unit area in unit time. The displacement that a macroscopic sensor is sensitive to, $\mathbf{U}$ say, is the total average within $V_A$; i.e., $\mathbf{U}=\phi\mathbf{u}_f+(1-\phi)\mathbf{u}_s=\mathbf{u}+\mathbf{w}$. The bulk-stress tensor $\boldsymbol{\tau}$ of Eqs. (3) and (6) is defined $\boldsymbol{\tau}=\phi\boldsymbol{\tau}_f+(1-\phi)\boldsymbol{\tau}_s$, where $\boldsymbol{\tau}_s$ and $\boldsymbol{\tau}_f$ are the average stress tensors in the grains and fluid, respectively. It is easily shown (e.g., Pride, 1994) that over frequencies where grain scattering is unimportant, the fluid supports no averaged shear deformation and the average fluid stress is purely isotropic; i.e., $\boldsymbol{\tau}_f=-p\mathbf{I}$, where $\mathbf{I}$ is the identity tensor and $p$ is the fluid pressure. The electromagnetic fields $\mathbf{E}$, $\mathbf{H}$, $\mathbf{D}$, and $\mathbf{B}$ represent their average value throughout all of $V_A$. It can be assumed that $\mathbf{E}$ and $\mathbf{B}$ are the fields that macroscopic antennas record. For the time-harmonic description being employed, all of these wave field variables are complex and frequency dependent.

If a source for mechanical waves exerts an average force density $\mathbf{f}_f$ on the fluid phase and $\mathbf{f}_s$ on the solid phase, then the source terms $\mathbf{F}$ and $\mathbf{f}$ of Eqs. (3)–(5) are defined $\mathbf{F}\equiv\phi\mathbf{f}_f+(1-\phi)\mathbf{f}_s$ and $\mathbf{f}\equiv\mathbf{f}_f$. For many exploration sources (explosions in particular) it can be assumed that the force applied to the fluid phase is the same as that applied to the solid; i.e., $\mathbf{f}_f=\mathbf{f}_s$ or $\mathbf{F}=\mathbf{f}$. However, in what follows, $\mathbf{F}$ and $\mathbf{f}$ will be treated distinctly. The primary source for the electromagnetic fields is the current-source density $\mathbf{C}$. A magnetic-source term is not included on the right-hand side of Faraday's law [Eq. (1)] because such a term is always due to loops of applied current; i.e., a magnetic source is exactly represented by a particular distribution of applied current $\mathbf{C}$ and is, thus not a fundamental (or necessary) quantity.

The transport tensors $\mathbf{S}$ (electrical conductivity), $\mathbf{L}$ (electrokinetic-coupling coefficient), and $\mathbf{K}$ (fluid-flow permeability) are complex and frequency dependent. In the generalized Darcy's law [Eq. (5)], $\eta$ is the fluid's shear viscosity. The porous-continuum statement of Onsager reciprocity requires that both $\mathbf{S}$ and $\mathbf{K}$ are symmetric; however, no such general constraints are placed on the symmetry of $\mathbf{L}$. Nonetheless, for the commonly assumed case in which the charge per unit area adsorbed to the grain surfaces is uniform throughout an averaging volume and, simultaneously, the diffuse-double layers are thin, one can show that $\mathbf{L}$ is symmetric and, indeed, possesses the same symmetry properties as $\mathbf{S}$ (this is seen by generalizing the arguments in Pride, 1994, to the anisotropic case). Throughout what follows, it will be assumed that $\mathbf{L}$ is symmetric. In the isotropic case we define $\mathbf{S}=\sigma\mathbf{I}$, $\mathbf{L}=L\mathbf{I}$, and $\mathbf{K}=k\mathbf{I}$. Analytical expressions relating the isotropic coefficients $\sigma(\omega)$, $L(\omega)$, and $k(\omega)$ to frequency, fluid chemistry, and pore geometry are given by Pride (1994) to which the reader is referred for more detail.

The fourth-order stiffness tensor ${}_4\mathscr{S}$ is expressed as ${}_4\mathscr{S}=\mathscr{S}_{ijkl}\hat{\mathbf{x}}_i\hat{\mathbf{x}}_j\hat{\mathbf{x}}_k\hat{\mathbf{x}}_l$ in polyadic notation where the unit base vectors $\hat{\mathbf{x}}_i$ may correspond to any curvilinear coordinate system; however, we assume from the outset that they are orthogonal ($\hat{\mathbf{x}}_i\cdot\hat{\mathbf{x}}_j=0$; $i\neq j$). Material independent symmetry properties of ${}_4\mathscr{S}$ are obtained, as usual, by the requirements that: (1) rigid-body rotations produce no stress ($\mathscr{S}_{ijkl}=\mathscr{S}_{ijlk}$); (2) no couple stresses are present ($\mathscr{S}_{jikl}=\mathscr{S}_{ijkl}$); and (3) there exists a strain energy function ($\mathscr{S}_{ijkl}=\mathscr{S}_{klij}$). As usual, these symmetries reduce the number of possible independent constants within ${}_4\mathscr{S}$ to 21. Any material symmetries will reduce this number further. The second-order cross-stiffness tensor $\mathscr{C}$ must also be symmetric ($\mathscr{C}_{ij}=\mathscr{C}_{ji}$) because no couple stresses are being allowed for, while the stiffness $M$ is a scalar. In the isotropic case, we define

$${}_4\mathscr{S}=[(H-2G)\delta_{ij}\delta_{kl}+G(\delta_{il}\delta_{jk}+\delta_{ik}\delta_{jl})]\hat{\mathbf{x}}_i\hat{\mathbf{x}}_j\hat{\mathbf{x}}_k\hat{\mathbf{x}}_l, \tag{10}$$

$$\mathscr{C}=C\mathbf{I}, \tag{11}$$

where expressions that relate $H$, $C$, and $M$ to the underlying fluid, solid, and drained-framework-of-grains elastic moduli are given by Pride *et al.* (1992) (and are equivalent to those of Biot and Willis, 1957) for the special case in which the material in an averaging volume has roughly uniform properties (e.g., if there is a mixture of sand and shale bodies within $V_A$, then the standard definitions for $H$, $C$, and $M$ no longer hold—see Berryman and Milton, 1991 and Berryman and Pride, 1996, for a discussion of such cases). The stiffness $G$ is the porous-material shear modulus. All of the above stiffnesses may be assumed complex and frequency dependent. The imaginary parts correspond to anelastic processes not directly related to macroscopic flux. In this work, we use the notation $c=c'+ic''$ to denote the separation into real and imaginary parts. It will be seen that the second law of thermodynamics requires that the principle components of ${}_4\mathscr{L}''$ as well as $M''$ must all be negative when there is an assumed $e^{-i\omega t}$ time dependence.

The symmetric, second-order tensors $\boldsymbol{\mu}$ and $\boldsymbol{\epsilon}$ are the magnetic permeability and electrical permittivity. For nearly all porous media of interest it can be assumed that $\boldsymbol{\mu}=\mu_0\mathbf{I}$ (iron, nickel, and cobalt are not significantly present in most sedimentary materials). In the isotropic case, we define $\boldsymbol{\epsilon}=\epsilon\mathbf{I}$. In general, $\boldsymbol{\epsilon}$ will be complex and frequency dependent. However, as discussed by Pride (1994), if $\epsilon_s/\epsilon_f \ll 1$ (where $\epsilon_s$ and $\epsilon_f$ are the real permittivities of the grains and fluid) and if electric double layers are thin relative to grain dimensions, then effectively no grain-scale ionic polarization will take place and $\boldsymbol{\epsilon}$ can be taken as real and frequency independent. These two conditions are usually met in porous media such as sedimentary rocks. Nonetheless, if there are losses associated with the processes of either electrical or magnetic polarization, it will be seen that the second law requires that the principal components of $\boldsymbol{\mu}''$ and $\boldsymbol{\epsilon}''$ must be positive.

Lastly, the bulk density $\rho$ in Eq. (3) is defined $\rho\equiv\phi\rho_f+(1-\phi)\rho_s$, where $\rho_f$ and $\rho_s$ are the purely real mass densities of the fluid and solid.

In the following sections we will derive a series of related results that are most easily obtained if the transport equation for fluid flux $\dot{\mathbf{w}}$ [Eq. (5)] is dot multiplied by $\eta\mathbf{K}^{-1}$ to give an effective force balance on the fluid in relative motion

$$\eta\mathbf{K}^{-1}\cdot\dot{\mathbf{w}}=-\nabla p+i\omega\rho_f\dot{\mathbf{u}}+\mathbf{f}+\eta(\mathbf{K}^{-1}\cdot\mathbf{L})\cdot\mathbf{E}. \tag{12}$$

The frame of reference for this force balance is fixed on the grains and is accelerating as $-i\omega\dot{\mathbf{u}}(\omega)$ thus giving rise to the apparent body force $i\omega\rho_f\dot{\mathbf{u}}(\omega)$. The inverse of a symmetric tensor is symmetric as is the dot product of two symmetric tensors. Thus $\mathbf{K}^{-1}\cdot\mathbf{L}$ is symmetric.

### B. The energy balance

We now obtain the porous-continuum energy balance using the complex wave fields defined above. From Eqs. (1) and (2), the following products are constructed where a star denotes the complex conjugate

$$\mathbf{H}^*\cdot[\nabla\times\mathbf{E}=i\omega\mathbf{B}], \tag{13}$$

$$\mathbf{E}\cdot[\nabla\times\mathbf{H}^*=i\omega\mathbf{D}^*+\mathbf{J}^*+\mathbf{C}^*]. \tag{14}$$

Subtracting Eq. (13) from (14) and using $\nabla\cdot(\mathbf{a}\times\mathbf{b})=\mathbf{b}\cdot\nabla\times\mathbf{a}-\mathbf{a}\cdot\nabla\times\mathbf{b}$ then gives

$$-\nabla\cdot(\mathbf{E}\times\mathbf{H}^*)=i\omega(\mathbf{E}\cdot\mathbf{D}^*-\mathbf{H}^*\cdot\mathbf{B})+\mathbf{E}\cdot\mathbf{J}^*+\mathbf{E}\cdot\mathbf{C}^*, \tag{15}$$

which is the complex-energy balance for the electromagnetic fields.

Similarly, from Eqs. (3) and (12), the following products are constructed

$$\dot{\mathbf{u}}^*\cdot[\nabla\cdot\boldsymbol{\tau}=-i\omega(\rho\dot{\mathbf{u}}+\rho_f\dot{\mathbf{w}})-\mathbf{F}], \tag{16}$$

$$\dot{\mathbf{w}}^*\cdot[-\nabla p=-i\omega\rho_f\dot{\mathbf{u}}-\mathbf{f}+\eta\mathbf{K}^{-1}\cdot\dot{\mathbf{w}}-\eta\mathbf{K}^{-1}\cdot\mathbf{L}\cdot\mathbf{E}]. \tag{17}$$

Combining with $\nabla\cdot(\mathbf{a}\cdot\mathbf{A})=\mathbf{A}:\nabla\mathbf{a}+(\nabla\cdot\mathbf{A}^T)\cdot\mathbf{a}$ and the fact that $\boldsymbol{\tau}=\boldsymbol{\tau}^T$ then gives

$$\begin{aligned}\nabla\cdot(\dot{\mathbf{u}}^*\cdot\boldsymbol{\tau}-\dot{\mathbf{w}}^*p)=i\omega[&\boldsymbol{\tau}:(\nabla\mathbf{u})^*-p(\nabla\cdot\mathbf{w})^*]\\&-i\omega[\rho|\dot{\mathbf{u}}|^2+\rho_f(\dot{\mathbf{u}}^*\cdot\dot{\mathbf{w}}+\dot{\mathbf{u}}\cdot\dot{\mathbf{w}}^*)]\\&+\dot{\mathbf{w}}^*\cdot\mathbf{g}-(\dot{\mathbf{u}}^*\cdot\mathbf{F}+\dot{\mathbf{w}}^*\cdot\mathbf{f}),\end{aligned} \tag{18}$$

where $\mathbf{g}$ is the mechanical force driving relative flow and is defined

$$\mathbf{g}=-\nabla p+i\omega\rho_f\dot{\mathbf{u}}+\mathbf{f}. \tag{19}$$

Equation (18) is the complex-energy balance for the mechanical fields.

We next add the electromagnetic and mechanical energy balances and separate into real and imaginary parts. This results in two purely real balances

$$-\nabla\cdot\mathbf{J}_e'=Q'+S_e', \tag{20}$$

$$-\nabla\cdot\mathbf{J}_e''=Q''+S_e'', \tag{21}$$

where $\mathbf{J}_e$ is the complex-energy flux and is defined

$$\mathbf{J}_e=\mathbf{E}\times\mathbf{H}^*-(\dot{\mathbf{u}}^*\cdot\boldsymbol{\tau}-\dot{\mathbf{w}}^*p), \tag{22}$$

while $S_e$ is the rate density at which the sources create complex energy and is defined

$$S_e=\mathbf{E}\cdot\mathbf{C}^*-(\dot{\mathbf{u}}^*\cdot\mathbf{F}+\dot{\mathbf{w}}^*\cdot\mathbf{f}). \tag{23}$$

The real vector $\mathbf{J}_e'$ represents the total amount of energy traversing unit area of the porous continuum in unit time while $S_e'$ represents the total amount of energy being created by the sources per unit time and volume. The real scalar $Q'$ is the rate at which energy is being irreversibly lost to heat per unit volume while $Q''$ is the rate at which energy is being reversibly stored per unit volume ($Q''$ is the reactive energy). Both $Q'$ and $Q''$ are defined by substituting the constitutive laws into the complex-energy balances to obtain

$$Q' + iQ'' = i\omega \boldsymbol{\epsilon}^* : \tfrac{1}{2}(\mathbf{EE}^* + \mathbf{E}^*\mathbf{E}) - i\omega \boldsymbol{\mu} : \tfrac{1}{2}(\mathbf{HH}^* + \mathbf{H}^*\mathbf{H})$$
$$+ i\omega {}_4\mathscr{L} :: \tfrac{1}{2}[(\nabla \mathbf{u})(\nabla \mathbf{u})^* + (\nabla \mathbf{u})^*(\nabla \mathbf{u})]$$
$$+ i\omega \mathscr{C} : [(\nabla \cdot \mathbf{w})(\nabla \mathbf{u})^* + (\nabla \cdot \mathbf{w})^*(\nabla \mathbf{u})]$$
$$+ i\omega M (\nabla \cdot \mathbf{w})^*(\nabla \cdot \mathbf{w}) - i\omega \rho |\dot{\mathbf{u}}|^2$$
$$- i\omega \rho_f (\dot{\mathbf{u}} \cdot \dot{\mathbf{w}}^* + \dot{\mathbf{u}}^* \cdot \dot{\mathbf{w}}) + \mathbf{S}^* : \tfrac{1}{2}(\mathbf{EE}^* + \mathbf{E}^*\mathbf{E})$$
$$+ \frac{\mathbf{K}^*}{\eta} : \tfrac{1}{2}(\mathbf{gg}^* + \mathbf{g}^*\mathbf{g}) + \mathbf{L}^* : (\mathbf{Eg}^* + \mathbf{E}^*\mathbf{g}). \tag{24}$$

In this expression, the factors involving the wave fields ($\mathbf{E}$, $\mathbf{H}$, $\mathbf{u}$, $\mathbf{w}$, and $\mathbf{g}$) are all purely real. Thus the separation into real and imaginary parts ($Q' + iQ''$) is defined entirely by the real and imaginary parts of the various constitutive coefficients. In writing Eq. (24) in this form, we have used the fact that $\mathbf{a}^* \cdot \mathbf{A} \cdot \mathbf{a} = \mathbf{A} : (\mathbf{aa}^* + \mathbf{a}^*\mathbf{a})/2$ for a symmetric tensor $\mathbf{A}$.

Equation (20) is the conservation of energy and is the central result of this section. Perhaps the main usefulness of Eq. (21) is that it helps provide physical insight into the meaning of the (real) vector $\mathbf{J}_e''$: away from the sources, any accumulation of the energy transported as $\mathbf{J}_e''$ does so reversibly and is associated with the difference between energy being stored in the fields (the potential energies) and the kinetic energies.

Finally, it is convenient to establish here conditions underwhich the second law of thermodynamics $Q' > 0$ is satisfied. From Eq. (24), $Q'$ may be written as

$$Q' = \omega \mathbf{H}^* \cdot \boldsymbol{\mu}'' \cdot \mathbf{H} + [\mathbf{E}^*, \mathbf{g}^*] \cdot \begin{bmatrix} \mathbf{S}' + \omega \boldsymbol{\epsilon}'' & \mathbf{L}' \\ \mathbf{L}' & \mathbf{K}' \end{bmatrix} \cdot \begin{bmatrix} \mathbf{E} \\ \mathbf{g} \end{bmatrix}$$
$$+ \omega [(\nabla \cdot \mathbf{w})^*, e_1^*, \ldots, e_6^*]$$
$$\cdot \begin{bmatrix} -M'' & -\mathscr{C}_1'' & \ldots & -\mathscr{C}_6'' \\ -\mathscr{C}_1'' & -\mathscr{L}_{11}'' & \ldots & -\mathscr{L}_{16}'' \\ \vdots & \vdots & \ddots & \vdots \\ -\mathscr{C}_6'' & -\mathscr{L}_{16}'' & \ldots & -\mathscr{L}_{66}'' \end{bmatrix} \cdot \begin{bmatrix} \nabla \cdot \mathbf{w} \\ e_1 \\ \vdots \\ e_6 \end{bmatrix}. \tag{25}$$

In the third term, the strains $e_\alpha$ and the moduli $\mathscr{C}_\alpha''$ and $\mathscr{L}_{\alpha\beta}''$ are related to their corresponding tensors by the usual mapping (e.g., Nye, 1985)

$$\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \rightarrow e_\alpha,$$
$$\mathscr{C}_{ij}'' \rightarrow \mathscr{C}_\alpha'', \quad \mathscr{L}_{ijkl}'' \rightarrow \mathscr{L}_{\alpha\beta}'', \tag{26}$$

where the $ij \rightarrow \alpha$ map is defined $11 \rightarrow 1$, $22 \rightarrow 2$, $33 \rightarrow 3$, $23 \rightarrow 4$, $13 \rightarrow 5$, and $12 \rightarrow 6$.

Each of the three terms comprising $Q'$ are in quadratic form (i.e., of the form $\mathbf{a}^* \cdot \mathbf{A} \cdot \mathbf{a}$). The second law requires that $Q' > 0$ unless all the macroscopic fields involved ($\mathbf{E}$, $\mathbf{H}$, $\mathbf{g}$, $\nabla \cdot \mathbf{w}$, and $e_\alpha$) are zero in which case $Q' = 0$. By considering the presence of arbitrary combinations of field types, we must have that each of the coefficient matrices in the three terms of Eq. (25) is positive definite [a real matrix $\mathbf{A}$ is called ''positive definite'' if $\mathbf{a}^* \cdot \mathbf{A} \cdot \mathbf{a} > 0$ for an arbitrary vector $\mathbf{a}$]. A positive-definite matrix has the property that each of its diagonal elements must be positive as well as all of its eigenvalues. More completely, the $N$ necessary and sufficient conditions required for an $N \times N$ matrix $A_{ij}$ to be positive definite can be written (e.g., Noble, 1969)

$$A_{11} > 0, \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} > 0, \ldots, \begin{vmatrix} A_{11} & \ldots & A_{1N} \\ \vdots & \ddots & \vdots \\ A_{N1} & \ldots & A_{NN} \end{vmatrix} > 0; \tag{27}$$

i.e., the determinant of the minors of $A_{ij}$ must all be positive. Each of the three coefficient matrices in Eq. (25) must satisfy constraints of this form in order to satisfy the second law. Thus the principal components of ${}_4\mathscr{L}''$ and $M''$ must all be negative while the principal components of $\mathbf{S}'$, $\mathbf{K}'$, $\boldsymbol{\epsilon}''$, and $\boldsymbol{\mu}''$ must all be positive. The positive-definite nature of these coefficient matrices will now be used in establishing uniqueness.

## C. Uniqueness

We next ask the following question: Within some volume $V$ of interest, is it possible to have two sets of fields—$\mathbf{u}_1$, $\mathbf{w}_1$, $\mathbf{E}_1$, $\mathbf{H}_1$ and $\mathbf{u}_2$, $\mathbf{w}_2$, $\mathbf{E}_2$, $\mathbf{H}_2$—that are both generated by the same sources and that both satisfy the same prescribed boundary conditions on the surface $S$ surrounding $V$ but that are otherwise different functions throughout $V$? To answer this, the difference fields

$$\delta \mathbf{u} = \mathbf{u}_1 - \mathbf{u}_2, \quad \delta \mathbf{w} = \mathbf{w}_1 - \mathbf{w}_2, \quad \delta \mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2,$$
$$\delta \mathbf{H} = \mathbf{H}_1 - \mathbf{H}_2, \tag{28}$$

are introduced. If the difference fields can be shown to be everywhere zero, the fields are said to be ''unique.''

The following products of difference fields may be constructed in analogy to Eqs. (13)–(17)

$$\delta \mathbf{H}^* \cdot [\nabla \times \delta \mathbf{E} = i\omega \boldsymbol{\mu} \cdot \delta \mathbf{H}], \tag{29}$$

$$\delta \mathbf{E} \cdot [\nabla \times \delta \mathbf{H}^* = i\omega \boldsymbol{\epsilon}^* \cdot \delta \mathbf{E}^* + \mathbf{S}^* \cdot \delta \mathbf{E}^* + \mathbf{L}^* \cdot \delta \mathbf{g}^*], \tag{30}$$

$$\delta \dot{\mathbf{u}}^* \cdot [\nabla \cdot \delta \boldsymbol{\tau} = -i\omega(\rho \delta \dot{\mathbf{u}} + \rho_f \delta \dot{\mathbf{w}})], \tag{31}$$

$$\delta \dot{\mathbf{w}}^* \cdot [-\nabla \delta p = -i\omega \rho_f \delta \dot{\mathbf{u}} + \eta \mathbf{K}^{-1} \cdot \delta \dot{\mathbf{w}} - \eta \mathbf{K}^{-1} \cdot \mathbf{L} \cdot \delta \mathbf{E}], \tag{32}$$

where the source terms are now missing since they are common to both sets of fields and where $\delta \mathbf{g} = -\nabla \delta p + i\omega \delta \dot{\mathbf{u}}$. The difference stresses $\delta \boldsymbol{\tau}$ and $-\delta p$ have the obvious definition from Eqs. (6) and (7). Going through the equivalent steps that led to the energy balance and then integrating over the volume $V$ and using the divergence theorem gives

$$\int_S \mathbf{n} \cdot \mathrm{Re}\{\delta \mathbf{E} \times \delta \mathbf{H}^* - \delta \boldsymbol{\tau} \cdot \delta \dot{\mathbf{u}}^* + \delta p \, \delta \dot{\mathbf{w}}^*\} dS = \int_V \delta Q' \, dV, \tag{33}$$

where

$$\delta Q' = \omega\,\delta\mathbf{H}^*\cdot\boldsymbol{\mu}''\cdot\delta\mathbf{H} + [\delta\mathbf{E}^*, \delta\mathbf{g}^*]\cdot\begin{bmatrix}\mathbf{S}'+\omega\boldsymbol{\epsilon}'' & \mathbf{L}' \\ \mathbf{L}' & \mathbf{K}'\end{bmatrix}\cdot\begin{bmatrix}\delta\mathbf{E}\\ \delta\mathbf{g}\end{bmatrix}$$

$$+\omega[(\nabla\cdot\delta\mathbf{w})^*, \delta e_1^*, \ldots, \delta e_6^*]$$

$$\cdot\begin{bmatrix} -M'' & -\mathscr{C}_1'' & \ldots & -\mathscr{C}_6'' \\ -\mathscr{C}_1'' & -\mathscr{L}_{11}'' & \ldots & -\mathscr{L}_{16}'' \\ \vdots & \vdots & \ddots & \vdots \\ -\mathscr{C}_6'' & -\mathscr{L}_{16}'' & \ldots & -\mathscr{L}_{66}'' \end{bmatrix}\cdot\begin{bmatrix}\nabla\cdot\delta\mathbf{w}\\ \delta e_1 \\ \vdots \\ \delta e_6\end{bmatrix}. \tag{34}$$

The difference strains $\delta e_\alpha$ are defined as in Eq. (26).

If it is now required that each of the following three types of boundary conditions are established on the surface $S$

(i) $\mathbf{n}\times\mathbf{H}$ or $\mathbf{n}\times\mathbf{E}$ specified,

(ii) $\mathbf{n}\cdot\boldsymbol{\tau}$ or $\mathbf{u}$ specified, (35)

(iii) $p$ or $\mathbf{n}\cdot\mathbf{w}$ specified,

then the surface integral in Eq. (33) vanishes. Because $V$ is arbitrary, we then must have

$$\delta Q' = 0 \tag{36}$$

everywhere within $V$. Because each of the coefficient matrices in the three quadratic forms comprising $\delta Q'$ are positive definite, we then must have

$$\delta\mathbf{E}=0,\quad \delta\mathbf{g}=0,\quad \delta\mathbf{H}=0,\quad \nabla\cdot\delta\mathbf{w}=0,\quad \delta e_\alpha=0. \tag{37}$$

From $\delta\mathbf{E}=0$ and $\delta\mathbf{g}=0$, Eq. (32) gives (after multiplying by $\mathbf{K}/\eta$ and using the definition of $\delta\mathbf{g}$) that $\delta\mathbf{w}=0$. From $\nabla\cdot\delta\mathbf{w}=0$ and $\delta e_\alpha=0$, we have that $\delta\boldsymbol{\tau}=0$ so that Eq. (31) gives $\delta\mathbf{u}=0$. Thus because all the difference fields are zero, the electroseismic problem possesses unique field solutions when the boundary conditions in Eqs. (35) are satisfied.

Under conditions where $\boldsymbol{\epsilon}''$, $\boldsymbol{\mu}''$, $M''$, $\mathscr{C}_\alpha''$, and $\mathscr{L}_{\alpha\beta}''$ are all identically zero (the only loss is due to macroscopic transport), the requirement $\delta Q'=0$ only gives that $\delta\mathbf{E}=0$ and $\delta\mathbf{g}=0$. However, Eq. (29) gives $\delta\mathbf{H}=0$, while Eq. (32) again gives $\delta\mathbf{w}=0$. Thus in this case, $\delta\mathbf{u}$ satisfies $\nabla\cdot({}_4\mathscr{L}:\nabla\delta\mathbf{u}) = -\omega^2\rho\delta\mathbf{u}$ which are the elastodynamic equations. It is well known (e.g., Love, 1944, Sec. 124) that the displacement field is unique in elastodynamics so long as 1) $\mathbf{n}\cdot\boldsymbol{\tau}$ or $\mathbf{u}$ are specified on $S$, and 2) at some initial time the displacements are known. Without this initial condition, the solid-displacement field is nonunique when the framework of grains is lossless. Thus when the conditions of Eq. (35) are satisfied and when there is some initial knowledge of the solid displacements (e.g., $\mathbf{u}=0$ at time zero), the electroseismic wave fields are always unique throughout $V$.

### D. Reciprocity

Define $\mathbf{E}_1$, $\mathbf{H}_1$, $\mathbf{u}_1$, and $\mathbf{w}_1$ as the fields generated by the distribution of sources $\mathbf{C}_1$, $\mathbf{F}_1$, and $\mathbf{f}_1$, and the fields $\mathbf{E}_2$, $\mathbf{H}_2$, $\mathbf{u}_2$, and $\mathbf{w}_2$ as the fields generated by $\mathbf{C}_2$, $\mathbf{F}_2$, and $\mathbf{f}_2$. All of these fields and sources exist in the same (possibly heterogeneous) porous material. Consider the following products involving the governing equations (which have been slightly rewritten)

$$\mathbf{H}_i\cdot[\nabla\times\mathbf{E}_j = i\omega\boldsymbol{\mu}\cdot\mathbf{H}_j], \tag{38}$$

$$\mathbf{E}_i\cdot[\nabla\times\mathbf{H}_j = -i\omega\mathbf{P}\cdot\mathbf{E}_j - \omega^2\mathbf{L}\cdot\mathbf{R}\cdot\mathbf{w}_j + \mathbf{C}_j], \tag{39}$$

$$-i\omega\mathbf{u}_i\cdot[\nabla\cdot\boldsymbol{\tau}_j = -\omega^2(\rho\mathbf{u}_j+\rho_f\mathbf{w}_j) - \mathbf{F}_j], \tag{40}$$

$$-i\omega\mathbf{w}_i\cdot[-\nabla p_j = -\omega^2(\rho_f\mathbf{u}_j + \mathbf{R}\cdot\mathbf{w}_j) + i\omega\mathbf{R}\cdot\mathbf{L}\cdot\mathbf{E}_j - \mathbf{f}_j], \tag{41}$$

where we have defined an effective complex inertia tensor $\mathbf{R}$ for the fluid in relative motion

$$\mathbf{R} \equiv \frac{i\eta}{\omega}\mathbf{K}^{-1}, \tag{42}$$

and an effective complex electrical-permittivity tensor $\mathbf{P}$ as

$$\mathbf{P} \equiv \boldsymbol{\epsilon} + \frac{i}{\omega}\mathbf{S} - \mathbf{L}\cdot\mathbf{R}\cdot\mathbf{L}. \tag{43}$$

Using $\mathbf{R}$ and $\mathbf{P}$ symbolically emphasizes the wave properties of the equations as opposed to the diffusive properties. Both $\mathbf{R}$ and $\mathbf{P}$ are symmetric (if $\mathbf{L}$ is taken as symmetric) so that $\mathbf{R}\cdot\mathbf{L}=\mathbf{L}\cdot\mathbf{R}$. The source indices $i$ and $j$ in Eqs. (38)–(41) equal 1 or 2 with $i\neq j$. Letting $i=1$ and $j=2$ and then $i=2$ and $j=1$, allows the following to be constructed by subtraction

$$\nabla\cdot(\mathbf{E}_1\times\mathbf{H}_2 - \mathbf{E}_2\times\mathbf{H}_1) = \omega^2(\mathbf{L}\cdot\mathbf{R}):(\mathbf{E}_1\mathbf{w}_2 - \mathbf{E}_2\mathbf{w}_1) - (\mathbf{E}_1\cdot\mathbf{C}_2 - \mathbf{E}_2\cdot\mathbf{C}_1), \tag{44}$$

$$-i\omega\nabla\cdot(\mathbf{u}_1\cdot\boldsymbol{\tau}_2 - \mathbf{u}_2\cdot\boldsymbol{\tau}_1) = -i\omega\mathscr{C}:[(\nabla\mathbf{u}_1)(\nabla\cdot\mathbf{w}_2) - (\nabla\mathbf{u}_2)(\nabla\cdot\mathbf{w}_1)] + i\omega^3\rho_f(\mathbf{u}_1\cdot\mathbf{w}_2 - \mathbf{u}_2\cdot\mathbf{w}_1) + i\omega(\mathbf{u}_1\cdot\mathbf{F}_2 - \mathbf{u}_2\cdot\mathbf{F}_1), \tag{45}$$

$$i\omega\nabla\cdot(\mathbf{w}_1 p_2 - \mathbf{w}_2 p_1) = i\omega\mathscr{C}:[(\nabla\mathbf{u}_1)(\nabla\cdot\mathbf{w}_2) - (\nabla\mathbf{u}_2)(\nabla\cdot\mathbf{w}_1)] - i\omega^3\rho_f(\mathbf{u}_1\cdot\mathbf{w}_2 - \mathbf{u}_2\cdot\mathbf{w}_1) - \omega^2(\mathbf{L}\cdot\mathbf{R}):(\mathbf{E}_1\mathbf{w}_2 - \mathbf{E}_2\mathbf{w}_1) + i\omega(\mathbf{w}_1\cdot\mathbf{f}_2 - \mathbf{w}_2\cdot\mathbf{f}_1). \tag{46}$$

Adding these equations and integrating over a volume $V$ of the material then gives

$$-\int_S \mathbf{n}\cdot[\mathbf{E}_1\times\mathbf{H}_2 - \mathbf{E}_2\times\mathbf{H}_1 - i\omega(\mathbf{u}_1\cdot\boldsymbol{\tau}_2 - \mathbf{u}_2\cdot\boldsymbol{\tau}_1 - \mathbf{w}_1 p_2 + \mathbf{w}_2 p_1)]\,dS = \int_V[\mathbf{E}_1\cdot\mathbf{C}_2 - \mathbf{E}_2\cdot\mathbf{C}_1 - i\omega(\mathbf{u}_1\cdot\mathbf{F}_2 - \mathbf{u}_2\cdot\mathbf{F}_1 + \mathbf{w}_1\cdot\mathbf{f}_2 - \mathbf{w}_2\cdot\mathbf{f}_1)]\,dV. \tag{47}$$

This is the general statement of reciprocity. If the sources $(\mathbf{C}_i, \mathbf{F}_i, \mathbf{f}_i)$ all have finite spatial extent, then the fields from these sources will necessarily fall off at distances greater than the source dimensions. Thus, for finite sources, if the volume is expanded so that the fields are completely negligible on $S$, the following statement of reciprocity is obtained

$$\int_{V_{C2}} \mathbf{E}_1 \cdot \mathbf{C}_2 dV - \int_{V_{C1}} \mathbf{E}_2 \cdot \mathbf{C}_1 \, dV - i\omega \left[ \int_{V_{F2}} \mathbf{u}_1 \cdot \mathbf{F}_2 dV - \int_{V_{F1}} \mathbf{u}_2 \cdot \mathbf{F}_1 \, dV \right] - i\omega \left[ \int_{V_{f2}} \mathbf{w}_1 \cdot \mathbf{f}_2 dV - \int_{V_{f1}} \mathbf{w}_2 \cdot \mathbf{f}_1 \, dV \right] = 0, \quad (48)$$

where the volumes $V_{C2}$, $V_{F2}$, etc. are defined to just enclose their respective source terms.

This statement is very useful especially when the source fields are compact enough to be treated as point sources. As a specific example, consider the following two point-source distributions

$$\begin{aligned} &\mathbf{F}_1(\mathbf{r}) = mNgs_1(\omega)\delta(\mathbf{r}-\mathbf{r}_1)\hat{\mathbf{z}}, \\ &\mathbf{f}_1(\mathbf{r}) = mNgs_1(\omega)\delta(\mathbf{r}-\mathbf{r}_1)\hat{\mathbf{z}}, \quad \mathbf{C}_1 = 0, \\ &\mathbf{F}_2(\mathbf{r}) = 0, \quad \mathbf{f}_2(\mathbf{r}) = 0, \\ &\mathbf{C}_2(\mathbf{r}) = -i\omega d s_2(\omega)\delta(\mathbf{r}-\mathbf{r}_2)\hat{\mathbf{x}}. \end{aligned} \quad (49)$$

The first distribution corresponds to a vertical impact source; i.e., a mass $m$ accelerating at $Ng$ ($g$=9.8 m/s$^2$) impacts at the point $\mathbf{r}_1$ in the vertical direction $\hat{\mathbf{z}}$ with a normalized spectra $s_1(\omega)$ ($s_1$ is unitless and is of order one in amplitude). The second distribution corresponds to that of an electric dipole of moment $d$, situated at position $\mathbf{r}_2$, oriented in the horizontal direction $\hat{\mathbf{x}}_1$, and possessing a normalized spectra $s_2(\omega)$. Reciprocity [Eq. (48)] then gives that

$$\hat{\mathbf{z}} \cdot [\mathbf{u}_2(\mathbf{r}_1) + \mathbf{w}_2(\mathbf{r}_1)] = \frac{s_2(\omega)}{s_1(\omega)} \frac{d}{mNg} \hat{\mathbf{x}} \cdot \mathbf{E}_1(\mathbf{r}_2). \quad (50)$$

If we know, for example, the horizontal component of the electric field recorded at position $\mathbf{r}_2$ due to a point impact at $\mathbf{r}_1$ [i.e., $\hat{\mathbf{x}} \cdot \mathbf{E}_1(\mathbf{r}_2)$], then we know automatically the vertical component of the displacement field recorded at position $\mathbf{r}_1$ due to a horizontal-dipole antenna at position $\mathbf{r}_2$ {i.e., $\hat{\mathbf{z}} \cdot [\mathbf{u}_2(\mathbf{r}_1) + \mathbf{w}_2(\mathbf{r}_1)]$}.

Continuing with this example, lets say that the impact source is a hammer weighing 4 kg accelerated at $2g$ onto the surface of the Earth. Seismic waves are generated that then traverse an interface at depth generating an electric field. For an interface at a few tens of meters depth, a field strength of $10^{-6}$ V/m is easily recorded at the Earth's surface—see Mikhailov and Haartsen (1996) for experimental details involving a hammer source and Thompson and Gist (1993) for details involving an explosion source. Furthermore, if a horizontal-dipole current source on the Earth's surface produces an electromagnetic field that traverses this same interface, we expect seismic waves to be generated. In order to detect a seismic displacement on the order of $10^{-7}$ m (a typical displacement recorded in exploration surveys), Eq. (50) tells us that the dipole moment must be on the order of 10 Cm. Although somewhat large compared to conventional dipole sources used in exploration geophysics, an antenna with a dipole moment on this order could easily be constructed. Thus, as first suggested by Thompson and Gist (1993), there is the distinct possibility that detectable seismic waves are generated as electromagnetic waves traverse interfaces at depth. This result is but one demonstration of the great utility of reciprocity. Another is given in the following section.

## E. Representation integrals

Following the general idea of Gangi (1970), we now use reciprocity to arrive at the representation integrals for $\mathbf{u}$, $\mathbf{w}$, and $\mathbf{E}$ from which $\boldsymbol{\tau}$, $p$, and $\mathbf{H}$ can be obtained by differentiation. The primary fields $\mathbf{u}$, $\mathbf{w}$, and $\mathbf{E}$ are to be represented (or determined) within a volume $V$ of material—possibly heterogeneous—which is bounded by a surface $S$. The fields are either due to sources $\mathbf{F}$, $\mathbf{f}$, and $\mathbf{C}$ distributed within $V$ or else are due to sources external to $V$ in which case all field types are assumed known over $S$.

Let us begin by defining three specific sets of point sources within $V$

$$\begin{array}{llll} F \text{ set:} & \mathbf{F}^F = \mathbf{s}^F \delta(\mathbf{r}-\mathbf{r}'), & \mathbf{f}^F = 0, & \mathbf{C}^F = 0; \\ f \text{ set:} & \mathbf{F}^f = 0, & \mathbf{f}^f = \mathbf{s}^f \delta(\mathbf{r}-\mathbf{r}'), & \mathbf{C}^f = 0; \\ C \text{ set:} & \mathbf{F}^C = 0, & \mathbf{f}^C = 0, & \mathbf{C}^C = \mathbf{s}^C \delta(\mathbf{r}-\mathbf{r}'). \end{array} \quad (51)$$

The vectors $\mathbf{s}^F$, $\mathbf{s}^f$, and $\mathbf{s}^C$ are assumed spatially constant but are otherwise arbitrary. Corresponding to each of these point-source sets there are produced the ''fundamental'' vectors $\mathbf{u}^\xi$, $\mathbf{w}^\xi$, $\mathbf{E}^\xi$, and $\mathbf{H}^\xi$ as well as the fundamental stress tensor $\boldsymbol{\tau}^\xi$ and the fundamental pressure $p^\xi$ where $\xi$=$F$, $f$, or $C$ depending on the point-source set.

The reciprocity statement is now used. Identifying our unknown fields $\mathbf{u}$, $\mathbf{w}$, $\mathbf{E}$, etc. with the fields ''1'' in Eq. (47) and our fundamental quantities $\mathbf{u}^\xi$, $\mathbf{w}^\xi$, $\mathbf{E}^\xi$, etc. with the fields ''2,'' then gives the first variant of the representation integrals

$$\mathbf{s}^\xi \cdot \mathbf{X}_\xi = \int_V \{\mathbf{E}^\xi \cdot \mathbf{C} - i\omega[\mathbf{u}^\xi \cdot \mathbf{F} + \mathbf{w}^\xi \cdot \mathbf{f}]\} dV - \int_S \mathbf{n} \cdot \{\mathbf{E} \times \mathbf{H}^\xi - \mathbf{E}^\xi \times \mathbf{H} - i\omega[\boldsymbol{\tau}^\xi \cdot \mathbf{u} - \boldsymbol{\tau} \cdot \mathbf{u}^\xi - \mathbf{w}p^\xi + \mathbf{w}^\xi p]\} dS, \quad (52)$$

where, for compactness of expression, we have temporarily redefined the independent wave fields as $\mathbf{X}_F \equiv -i\omega\mathbf{u}$, $\mathbf{X}_f \equiv -i\omega\mathbf{w}$, and $\mathbf{X}_C \equiv \mathbf{E}$. Next, the Green's tensors are introduced as the linear proportionalities between the point-source directions $\mathbf{s}^\xi$ and the fundamental vectors

$$\mathbf{u}^\xi(\mathbf{r}|\mathbf{r}';\mathbf{s}^\xi) = \mathbf{G}_u^\xi(\mathbf{r}|\mathbf{r}') \cdot \mathbf{s}^\xi, \quad (53)$$

$$\mathbf{w}^\xi(\mathbf{r}|\mathbf{r}';\mathbf{s}^\xi) = \mathbf{G}_w^\xi(\mathbf{r}|\mathbf{r}') \cdot \mathbf{s}^\xi, \quad (54)$$

$$\mathbf{E}^\xi(\mathbf{r}|\mathbf{r}';\mathbf{s}^\xi) = \mathbf{G}_E^\xi(\mathbf{r}|\mathbf{r}') \cdot \mathbf{s}^\xi, \quad (55)$$

where again $\xi$=$F$, $f$, or $C$. Thus there are a total of nine primary Green's tensors involved in electroseismic theory. The Green's tensors do not depend on the arbitrary source vectors $\mathbf{s}^\xi$. The notation $\mathbf{r}|\mathbf{r}'$ is indicating, as usual, the response at point $\mathbf{r}$ due to a point source at $\mathbf{r}'$. Because the $\mathbf{s}^\xi$ are spatially constant, when Eqs. (53)–(55) are introduced into the governing equations, the following tensor equations are obtained

$$\nabla\times\mathbf{G}_E^{\xi}=i\omega\boldsymbol{\mu}\cdot\mathbf{G}_H^{\xi}, \tag{56}$$

$$\nabla\times\mathbf{G}_H^{\xi}=-i\omega\mathbf{P}\cdot\mathbf{G}_E^{\xi}-\omega^2(\mathbf{L}\cdot\mathbf{R})\cdot\mathbf{G}_w^{\xi}+\delta_{C\xi}\mathbf{I}\delta(\mathbf{r}-\mathbf{r}'), \tag{57}$$

$$\nabla\cdot{}_3\mathbf{T}^{\xi}=-\omega^2(\rho\mathbf{G}_u^{\xi}+\rho_f\mathbf{G}_w^{\xi})-\delta_{F\xi}\mathbf{I}\delta(\mathbf{r}-\mathbf{r}'), \tag{58}$$

$$-\nabla\mathbf{p}^{\xi}=-\omega^2(\rho_f\mathbf{G}_u^{\xi}+\mathbf{R}\cdot\mathbf{G}_w^{\xi})+i\omega(\mathbf{R}\cdot\mathbf{L})\cdot\mathbf{G}_E^{\xi} -\delta_{f\xi}\mathbf{I}\delta(\mathbf{r}-\mathbf{r}'), \tag{59}$$

where the third-order Green's bulk-stress tensor ${}_3\mathbf{T}^{\xi}$ and the Green's fluid-pressure vector $\mathbf{p}^{\xi}$ are related to the primary tensors as

$${}_3\mathbf{T}^{\xi}={}_4\mathscr{L}:\nabla\mathbf{G}_u^{\xi}+\mathscr{C}\ \nabla\cdot\mathbf{G}_w^{\xi} \tag{60}$$

$$-\mathbf{p}^{\xi}=\mathscr{C}:\nabla\mathbf{G}_u^{\xi}+M\nabla\cdot\mathbf{G}_w^{\xi}. \tag{61}$$

The $\delta_{C\xi}$, $\delta_{F\xi}$, and $\delta_{f\xi}$ are Kronecker delta functions. Equations (56)–(61) are the defining differential equations for the Green's tensors.

We now establish the reciprocal properties of the primary Green's tensors. Using Eq. (48), one can establish relations between the fundamental vectors such as

$$\int_{V_{F2}}[\mathbf{u}^F(\mathbf{r}|\mathbf{r}_1;\mathbf{s}_1^F)\cdot\mathbf{s}_2^F]\delta(\mathbf{r}-\mathbf{r}_2)d^3\mathbf{r} =\int_{V_{F1}}[\mathbf{u}^F(\mathbf{r}|\mathbf{r}_2;\mathbf{s}_2^F)\cdot\mathbf{s}_1^F]\delta(\mathbf{r}-\mathbf{r}_1)d^3\mathbf{r}, \tag{62}$$

which gives

$$\mathbf{s}_2^F\cdot\mathbf{u}^F(\mathbf{r}_2|\mathbf{r}_1;\mathbf{s}_1^F)=\mathbf{s}_1^F\cdot\mathbf{u}^F(\mathbf{r}_1|\mathbf{r}_2;\mathbf{s}_2^F). \tag{63}$$

The following identities can now be established where $\hat{\mathbf{x}}_i$ is the unit vector along the $i$th coordinate axis and where summation over the repeated indices is assumed

$$\begin{aligned}\mathbf{G}_u^F(\mathbf{r}_1|\mathbf{r}_2)&=\hat{\mathbf{x}}_i[\hat{\mathbf{x}}_i\cdot\mathbf{G}_u^F(\mathbf{r}_1|\mathbf{r}_2)\cdot\hat{\mathbf{x}}_j]\hat{\mathbf{x}}_j\\ &=\hat{\mathbf{x}}_i[\hat{\mathbf{x}}_i\cdot\mathbf{u}^F(\mathbf{r}_1|\mathbf{r}_2;\hat{\mathbf{x}}_j)]\hat{\mathbf{x}}_j\\ &=\hat{\mathbf{x}}_i[\hat{\mathbf{x}}_j\cdot\mathbf{u}^F(\mathbf{r}_2|\mathbf{r}_1;\hat{\mathbf{x}}_i)]\hat{\mathbf{x}}_j\\ &=\hat{\mathbf{x}}_i[\hat{\mathbf{x}}_j\cdot\mathbf{G}_u^F(\mathbf{r}_2|\mathbf{r}_1)\cdot\hat{\mathbf{x}}_i]\hat{\mathbf{x}}_j\\ &=[\mathbf{G}_u^F(\mathbf{r}_2|\mathbf{r}_1)]^T.\end{aligned} \tag{64}$$

Equation (63) was used between the second and third lines. Identical manipulations on the other fundamental vectors gives

$$\mathbf{G}_u^f(\mathbf{r}_1|\mathbf{r}_2)=[\mathbf{G}_w^F(\mathbf{r}_2|\mathbf{r}_1)]^T, \tag{65}$$

$$-i\omega\mathbf{G}_u^C(\mathbf{r}_1|\mathbf{r}_2)=[\mathbf{G}_E^F(\mathbf{r}_2|\mathbf{r}_1)]^T, \tag{66}$$

$$\mathbf{G}_w^f(\mathbf{r}_1|\mathbf{r}_2)=[\mathbf{G}_w^f(\mathbf{r}_2|\mathbf{r}_1)]^T, \tag{67}$$

$$-i\omega\mathbf{G}_w^C(\mathbf{r}_1|\mathbf{r}_2)=[\mathbf{G}_E^f(\mathbf{r}_2|\mathbf{r}_1)]^T, \tag{68}$$

$$\mathbf{G}_E^C(\mathbf{r}_1|\mathbf{r}_2)=[\mathbf{G}_E^C(\mathbf{r}_2|\mathbf{r}_1)]^T. \tag{69}$$

These define the general reciprocal (symmetry) properties of the Green's tensors.

Using these results in Eq. (52) then gives the representation integrals in final form

$$\mathbf{u}(\mathbf{r})=\Sigma_F(\mathbf{r})+\int_V\{\mathbf{G}_u^C(\mathbf{r}|\mathbf{r}')\cdot\mathbf{C}(\mathbf{r}')+\mathbf{G}_u^F(\mathbf{r}|\mathbf{r}')\cdot\mathbf{F}(\mathbf{r}') +\mathbf{G}_u^f(\mathbf{r}|\mathbf{r}')\cdot\mathbf{f}(\mathbf{r}')\}d^3\mathbf{r}', \tag{70}$$

$$\mathbf{w}(\mathbf{r})=\Sigma_f(\mathbf{r})+\int_V\{\mathbf{G}_w^C(\mathbf{r}|\mathbf{r}')\cdot\mathbf{C}(\mathbf{r}')+\mathbf{G}_w^F(\mathbf{r}|\mathbf{r}')\cdot\mathbf{F}(\mathbf{r}') +\mathbf{G}_w^f(\mathbf{r}|\mathbf{r}')\cdot\mathbf{f}(\mathbf{r}')\}d^3\mathbf{r}', \tag{71}$$

$$\mathbf{E}(\mathbf{r})=-i\omega\Sigma_C(\mathbf{r})+\int_V\{\mathbf{G}_E^C(\mathbf{r}|\mathbf{r}')\cdot\mathbf{C}(\mathbf{r}') +\mathbf{G}_E^F(\mathbf{r}|\mathbf{r}')\cdot\mathbf{F}(\mathbf{r}')+\mathbf{G}_E^f(\mathbf{r}|\mathbf{r}')\cdot\mathbf{f}(\mathbf{r}')\}d^3\mathbf{r}', \tag{72}$$

where the vectors $\Sigma_\xi$ ($\xi$=$F$, $f$, or $C$) are the surface-integral contributions and can be written

$$\begin{aligned}\Sigma_\xi(\mathbf{r})=\int_S\Bigg\{&\frac{1}{i\omega}[[\mathbf{G}_E^{\xi}(\mathbf{r}'|\mathbf{r})]^T\times\mathbf{n}]\cdot\mathbf{H}(\mathbf{r}')\\ &-[(\omega^2\boldsymbol{\mu})^{-1}\cdot[\nabla'\times\mathbf{G}_E^{\xi}(\mathbf{r}'\mathbf{r})]^T\times\mathbf{n}]\cdot\mathbf{E}(\mathbf{r}')\\ &-\mathbf{n}\cdot[{}_4\mathscr{L}:\nabla'[\mathbf{G}_u^{\xi}(\mathbf{r}'|\mathbf{r})]^T\\ &+\mathscr{C}\ \nabla'\cdot[\mathbf{G}_w^{\xi}(\mathbf{r}'|\mathbf{r})]^T]\cdot\mathbf{u}(\mathbf{r}')\\ &+[\mathscr{C}:\nabla'\mathbf{G}_u^{\xi}(\mathbf{r}'|\mathbf{r})\mathbf{n}+M\nabla'\cdot\mathbf{G}_w^{\xi}(\mathbf{r}'|\mathbf{r})\mathbf{n}]\cdot\mathbf{w}(\mathbf{r}')\\ &+[\mathbf{G}_u^{\xi}(\mathbf{r}'|\mathbf{r})\cdot\mathbf{n}]\cdot\boldsymbol{\tau}(\mathbf{r}')-[\mathbf{n}\cdot\mathbf{G}_w^{\xi}(\mathbf{r}'|\mathbf{r})]p(\mathbf{r}')\Bigg\}d^2\mathbf{r}'.\end{aligned} \tag{73}$$

Within these surface integrals, we have that $\mathbf{n}$, $\boldsymbol{\mu}$, ${}_4\mathbf{L}$, $\mathbf{C}$, and $M$ are all possible functions of $\mathbf{r}'$. Using the symmetry relations [Eqs. (64)–(69)], it is possible to express all of the Green's tensors within the surface integrals as functions of $\mathbf{r}|\mathbf{r}'$ (instead of $\mathbf{r}'|\mathbf{r}$ as given). Note that only the nine primary Green's tensors defined as $\mathbf{G}_u^{\xi}(\mathbf{r}|\mathbf{r}')$, $\mathbf{G}_w^{\xi}(\mathbf{r}|\mathbf{r}')$, and $\mathbf{G}_E^{\xi}(\mathbf{r}|\mathbf{r}')$, where $\xi$=$F$, $f$, or $C$ are required in the representation integrals. If analytical solutions are sought, then only six of these nine ($\mathbf{G}_u^F$, $\mathbf{G}_w^F$, $\mathbf{G}_E^F$, $\mathbf{G}_w^f$, $\mathbf{G}_E^f$, and $\mathbf{G}_E^C$) need be determined from Eqs. (56)–(61) because $\mathbf{G}_u^f$, $\mathbf{G}_u^C$, and $\mathbf{G}_w^C$ are obtainable from the symmetry relations.

## II. WAVES IN AN ISOTROPIC AND HOMOGENEOUS WHOLESPACE

The equations will now be solved for the simple case of a homogeneous and isotropic wholespace. In this case, the complete set of governing equations can be written

$$[(H-G)\nabla\nabla+(G\nabla^2+\omega^2\rho)\mathbf{I}]\cdot\mathbf{u}+[C\nabla\nabla+\omega^2\rho_f\mathbf{I}]\cdot\mathbf{w} =-\mathbf{F}, \tag{74}$$

$$[C\nabla\nabla+\omega^2\rho_f\mathbf{I}]\cdot\mathbf{u}+[M\nabla\nabla+\omega^2\tilde{\rho}\ \mathbf{I}]\cdot\mathbf{w}-i\omega\tilde{\rho}\ L\mathbf{E}=-\mathbf{f}, \tag{75}$$

$$[\nabla\nabla-(\nabla^2+\omega^2\mu\tilde{\epsilon})\ \mathbf{I}]\cdot\mathbf{E}+i\omega^3\mu\tilde{\rho}\ L\mathbf{w}=i\omega\mu\mathbf{C}, \tag{76}$$

where

$$\tilde{\rho}(\omega)=\frac{i}{\omega}\frac{\eta}{k(\omega)} \tag{77}$$

is the effective density of the fluid in relative motion and where

$$\widetilde{\epsilon}(\omega)=\epsilon(\omega)+\frac{i}{\omega}\,\sigma(\omega)-\widetilde{\rho}(\omega)L^2(\omega) \tag{78}$$

is the effective electrical permittivity of the porous continuum. Both the plane-wave and the point-source responses of Eqs. (74)–(76) will be obtained.

### A. Plane waves

Setting **C**, **F**, and **f** to zero and working with the plane-wave representations given by

$$\mathbf{u}=\mathscr{U}\,\exp(i\mathbf{k}\cdot\mathbf{r})\hat{\mathbf{u}}, \tag{79}$$

$$\mathbf{w}=\mathscr{W}\,\exp(i\mathbf{k}\cdot\mathbf{r})\hat{\mathbf{w}}, \tag{80}$$

$$\mathbf{E}=\mathscr{E}\,\exp(i\mathbf{k}\cdot\mathbf{r})\hat{\mathbf{e}}, \tag{81}$$

where $\hat{\mathbf{u}}$, $\hat{\mathbf{w}}$, and $\hat{\mathbf{e}}$ are unit vectors and where the wave vector $\mathbf{k}$ may be written

$$\mathbf{k}=\omega s(\omega)\hat{\mathbf{k}}, \tag{82}$$

with $s(\omega)$ the complex slowness (to be determined) and $\hat{\mathbf{k}}$ a unit vector in the direction of propagation, allows the governing equations to be written as

$$\begin{bmatrix} (H-G)s^2(\hat{\mathbf{k}}\cdot\hat{\mathbf{u}})\hat{\mathbf{k}}-(\rho-Gs^2)\hat{\mathbf{u}} & Cs^2(\hat{\mathbf{k}}\cdot\hat{\mathbf{w}})\hat{\mathbf{k}}-\rho_f\hat{\mathbf{w}} & 0 \\ Cs^2(\hat{\mathbf{k}}\cdot\hat{\mathbf{u}})\hat{\mathbf{k}}-\rho_f\hat{\mathbf{u}} & Ms^2(\hat{\mathbf{k}}\cdot\hat{\mathbf{w}})\hat{\mathbf{k}}-\widetilde{\rho}\hat{\mathbf{w}} & \frac{i}{\omega}\,\widetilde{\rho}\,L\hat{\mathbf{e}} \\ 0 & -i\omega\mu\widetilde{\rho}\,L\hat{\mathbf{w}} & s^2(\hat{\mathbf{k}}\cdot\hat{\mathbf{e}})\hat{\mathbf{k}}+(\widetilde{\epsilon}\mu-s^2)\hat{\mathbf{e}} \end{bmatrix}\begin{bmatrix}\mathscr{U}\\ \mathscr{W}\\ \mathscr{E}\end{bmatrix}=0. \tag{83}$$

We will consider in this section the various transverse and longitudinal wave modes associated with these equations.

By definition, a transverse mode is one in which the material response is in a direction perpendicular (transverse) to the direction of propagation (i.e., $\hat{\mathbf{k}}\cdot\hat{\mathbf{u}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{w}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{e}}=0$). Similarly, a longitudinal mode is one in which the material response and the wave direction are parallel (i.e., $\hat{\mathbf{k}}\cdot\hat{\mathbf{u}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{w}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{e}}=1$). It is easily shown from Eq. (83) that wave modes with mixed response (e.g., $\hat{\mathbf{k}}\cdot\hat{\mathbf{u}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{w}}=1$ but $\hat{\mathbf{k}}\cdot\hat{\mathbf{e}}=0$, etc.) lead to the trivial solution $\mathscr{U}=\mathscr{W}=\mathscr{E}=\mathrm{O}$ and, thus do not exist. Furthermore, one obtains that for both transverse and longitudinal modes $\hat{\mathbf{u}}=\hat{\mathbf{w}}=\hat{\mathbf{e}}$.

In what follows, to obtain the real phase velocity $v$ and the real attenuation coefficient $\alpha$ (units of inverse length) from the complex slownesses $s$, one must use the expressions

$$v(\omega)=1/\mathrm{Re}\{s(\omega)\}, \tag{84}$$

$$\alpha(\omega)=\omega\ \mathrm{Im}\{s(\omega)\}. \tag{85}$$

#### *1. Transverse modes*

Upon putting $\hat{\mathbf{k}}\cdot\hat{\mathbf{u}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{w}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{e}}=0$ and $\hat{\mathbf{u}}=\hat{\mathbf{w}}=\hat{\mathbf{e}}$ in Eq. (83), the transverse-mode system can be written

$$\begin{bmatrix} \rho-Gs^2 & \rho_f & 0 \\ \rho_f & \widetilde{\rho} & -\frac{i}{\omega}\,\widetilde{\rho}\,L \\ 0 & -i\omega\mu\widetilde{\rho}\,L & \widetilde{\epsilon}\mu-s^2 \end{bmatrix}\begin{bmatrix}\mathscr{U}\\ \mathscr{W}\\ \mathscr{E}\end{bmatrix}=0. \tag{86}$$

Nontrivial solutions for $\mathscr{U}$, $\mathscr{W}$, and $\mathscr{E}$ are obtained only if the determinant of the matrix vanishes. Upon taking the determinant, there are found two distinct complex transverse-wave slownesses

$$2s_{s,em}^2=\frac{\rho_t}{G}+\mu\widetilde{\epsilon}\left(1+\frac{\widetilde{\rho}\,L^2}{\widetilde{\epsilon}}\right) \pm\left\{\left[\frac{\rho_t}{G}-\mu\widetilde{\epsilon}\left(1+\frac{\widetilde{\rho}\,L^2}{\widetilde{\epsilon}}\right)\right]^2-4\mu\,\frac{\rho_f^2L^2}{G}\right\}^{1/2}, \tag{87}$$

where the $+$ in front of the radical corresponds to the mechanical shear-wave slowness $s_s$ and the $-$ corresponds to the electromagnetic (EM) slowness $s_{em}$. The complex density $\rho_t$ is defined

$$\rho_t=\rho-\rho_f^2/\widetilde{\rho}. \tag{88}$$

Associating the positive square root with the shear wave (and the negative with the EM wave) comes from considering the case where there is no electrokinetic coupling ($L=0$) in which case $s_s$ and $s_{em}$ take on their usual definitions. The allowance for $L$ makes essentially no difference for these transverse-wave slownesses but we will see that it can have a non-negligible effect on the compressional-wave slownesses.

In terms of the amplitude $\mathscr{U}$, Eq. (86) gives

$$\mathscr{W}=\frac{G}{\rho_f}\left(s^2-\frac{\rho}{G}\right)\mathscr{U}, \tag{89}$$

$$\mathscr{E}=-i\omega\mu\widetilde{\rho}\,L\,\frac{G}{\rho_f}\left(\frac{s^2-\rho/G}{s^2-\mu\widetilde{\epsilon}}\right)\mathscr{U}. \tag{90}$$

When $s=s_s$, these amplitudes correspond to that of a plane shear wave, while when $s=s_{em}$, they correspond to a plane EM wave.

There are always two polarizations possible for plane transverse waves. If the plane wave propagates in the $x$-$z$ plane of a Cartesian system $(x,y,z)$, then one polarization corresponds to $\hat{\mathbf{u}}=\hat{\mathbf{w}}=\hat{\mathbf{e}}=\hat{\mathbf{y}}$ and is the $SH-TE$ polarization (horizontally polarized displacement of shear waves ''*SH*'' and horizontally polarized electric field of EM waves ''*TE*''). The other polarization is given by $\hat{\mathbf{u}}=\hat{\mathbf{w}}=\hat{\mathbf{e}}=\hat{\mathbf{y}}\times\hat{\mathbf{k}}$

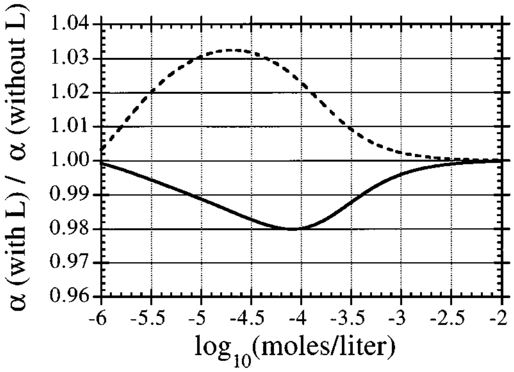


FIG. 1. The effect of the electrokinetic coupling coefficient $L$ on the compressional wave attenuation coefficients as a function of salinity of the saturating electrolyte. The dashed line is the slow-compressional wave while the solid line is the fast-compressional wave.

and is the $SV$–$TM$ polarization (vertically polarized displacements of shear waves ''$SV$'' and horizontally polarized magnetic field of EM waves ''$TM$'').

### 2. Longitudinal modes

Putting $\hat{\mathbf{u}}=\hat{\mathbf{w}}=\hat{\mathbf{e}}=\hat{\mathbf{k}}$ in Eq. (83) gives the system

$$\begin{bmatrix} Hs^2-\rho & Cs^2-\rho_f & 0 \\ Cs^2-\rho_f & Ms^2-\tilde{\rho} & \frac{i}{\omega}\tilde{\rho}\,L \\ 0 & i\omega\mu\tilde{\rho}\,L & -\mu\tilde{\epsilon} \end{bmatrix} \begin{bmatrix} \mathscr{U} \\ \mathscr{W} \\ \mathscr{E} \end{bmatrix} = 0. \tag{91}$$

Taking the determinant and setting it to zero gives two distinct complex longitudinal-wave slownesses

$$2s^2_{pf,ps} = \gamma \mp \sqrt{\gamma^2 - \frac{4\tilde{\rho}\rho}{MH-C^2}\left(\frac{\rho_t}{\rho}+\frac{\tilde{\rho}\,L^2}{\tilde{\epsilon}}\right)}, \tag{92}$$

where the $-$ corresponds to Biot's ''fast''-compressional-wave slowness $s_{pf}$ while the $+$ corresponds to the ''slow''-compressional-wave slowness $s_{ps}$ and where

$$\gamma = \frac{\rho M + \tilde{\rho}\,H(1+\tilde{\rho}\,L^2/\tilde{\epsilon}) - 2\rho_f C}{HM - C^2}. \tag{93}$$

As $L \rightarrow 0$ (no electrokinetic coupling), these slownesses correspond exactly to those defined by Biot (1956).

In terms of the amplitude $\mathscr{U}$, we then have

$$\mathscr{W} = -\left(\frac{Hs^2-\rho}{Cs^2-\rho_f}\right)\mathscr{U}, \tag{94}$$

$$\mathscr{E} = -\frac{i\omega\tilde{\rho}\,L}{\tilde{\epsilon}}\left(\frac{Hs^2-\rho}{Cs^2-\rho_f}\right)\mathscr{U}. \tag{95}$$

When $s=s_{pf}$, these amplitudes correspond to a fast-compressional wave, while when $s=s_{ps}$, they correspond to a slow-compressional wave.

As is seen in Fig. 1, the presence of the coupling coefficient $L$ can have a slight influence on the compressional-attenuation coefficients (on the order of a few percent). The effect on the phase velocity is less than one percent for this example. The material properties used in Fig. 1 correspond to a well-consolidated sandstone with 10-mD permeability and electrokinetic properties corresponding to quartz. Expressions for the various terms in Eq. (92) are given by Pride (1994) and the electrokinetic properties are consistent with the data of Li and de Bruyn (1966). The wave frequency is 100 Hz. We now interpret Fig. 1.

In the fast wave, the deviations in fluid pressure are generated by the frame compressions associated with the wave—regions of enhanced fluid pressure are generated by a frame contraction while regions of decreased fluid pressure are generated by a frame expansion. The frame compression and fluid compression occur in phase in the fast wave. These half-wavelength scale fluid-pressure gradients generate a fluid flow that carries along the excess counterions of the diffuse-double layer; i.e., the wave generates a streaming current. The counterions accumulate in the low-fluid-pressure regions and are depleted in the high-fluid-pressure regions thus generating an electric field. This electric field drives a conduction current that balances the streaming current so that the final degree of charge separation is controlled by both the electrokinetic and conductivity properties of the material. The generated electric field acts as a body force on the counterions in the diffuse-double layer that opposes the pressure gradient and, thus reduces the amount of relative flow. Therefore, in the fast wave, losses due to viscous shearing are reduced due to the electrokinetic effects as is seen in Fig. 1.

In the slow wave, the deviations in fluid pressure are generated directly from the fluid compressions associated with this wave. A region of high-fluid pressure generates a frame expansion while a region of low-fluid pressure generates a frame contraction. The frame compression and fluid compression thus occur out of phase in the slow wave resulting in the high fluid-pressure regions having an accumulation of counter charge and the low-fluid-pressure regions having a depletion (opposite from the fast wave). The fluid-pressure gradients will drive a relative flow that tends to reduce these charge separations (and pressure differences); however, the sign of the charge accumulations will not be reversed by this flow. Thus in the slow wave, the electric field acts as a body force that enhances the relative flow driven by the pressure gradient and results in a larger amount of viscous shearing as is seen in Fig. 1.

For both waves, the dependence on salinity is explained as follows. Both the width and the excess-charge content of the diffuse-double layers grow with decreasing salinity while the conductivity decreases. Thus as the salinity drops the amount of charge separation increases. However, when the salinity has decreased to values where surface conduction in the diffuse layer becomes important, the magnitude of the charge separation is decreased and the effect on the relative flow and attenuation is reduced all of which is seen in Fig. 1.

### 3. The eigen response

We now summarize the above for the case of plane waves in the $x$-$z$ plane of a Cartesian system $(x,y,z)$. The plane-wave direction is characterized as

$$\hat{\mathbf{k}}=[\sin\ \theta,\quad 0,\quad \cos\ \theta]^T \tag{96}$$

$$=[p/s,\quad 0,\quad q/s]^T, \tag{97}$$

where $\theta$ measures the direction of the wave relative to the positive $\hat{\mathbf{z}}$ direction, $p$ is the ''horizontal'' slowness, $q$ is the ''vertical'' slowness, $s$ is the appropriate plane-wave slowness, and $p^2+q^2=s^2$. In what follows, $p$ will be used to characterize wave direction rather than $\theta$. Note that in general, either $p$ or $q$ (or both) are complex because $s$ is complex.

For the $SH-TE$ transverse waves, the response is

$$\mathbf{u}=\hat{\mathbf{y}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SH-TE}, \tag{98}$$

$$\mathbf{w}=\frac{G}{\rho_f}\left(s^2-\frac{\rho}{G}\right)\hat{\mathbf{y}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SH-TE}, \tag{99}$$

$$\boldsymbol{\tau}=i\omega sG[\hat{\mathbf{k}}\hat{\mathbf{y}}+\hat{\mathbf{y}}\hat{\mathbf{k}}]e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SH-TE}, \tag{100}$$

$$p=0, \tag{101}$$

$$\mathbf{E}=i\omega\mu\ \frac{\tilde{\rho}}{\rho_f}LG\beta_T\hat{\mathbf{y}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SH-TE} \tag{102}$$

$$\mathbf{H}=i\omega s\ \frac{\tilde{\rho}}{\rho_f}LG\beta_T\hat{\mathbf{k}}\times\hat{\mathbf{y}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SH-TE}, \tag{103}$$

where

$$\beta_T=-\frac{s^2-\rho/G}{s^2-\mu\tilde{\epsilon}}, \tag{104}$$

$$\hat{\mathbf{k}}\hat{\mathbf{y}}+\hat{\mathbf{y}}\hat{\mathbf{k}}=\begin{bmatrix} 0 & p/s & 0 \\ p/s & 0 & q/s \\ 0 & q/s & 0 \end{bmatrix} \tag{105}$$

$$\hat{\mathbf{k}}\times\hat{\mathbf{y}}=[-q/s,0,\ p/s]^T, \tag{106}$$

$$\mathbf{k}\cdot\mathbf{r}=\omega(px+qz), \tag{107}$$

and where $s=s_s$ for $SH$ waves and $s=s_{em}$ for $TE$ waves. All responses have been normalized by a scale factor $\mathscr{S}^{SH-TE}$ that we are free to set (i.e., we have put $1/\mathscr{U}=\mathscr{S}^{SH-TE}$). Useful values for such scale factors will be discussed shortly. The plane-wave response is only known to within an arbitrary scale factor because the source amplitude is not fixed (i.e., the conditions for uniqueness, as discussed earlier, are being violated).

Similarly, for the $SV-TM$ transverse waves, the response is

$$\mathbf{u}=(\hat{\mathbf{y}}\times\hat{\mathbf{k}})e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SV-TM}, \tag{108}$$

$$\mathbf{w}=\frac{G}{\rho_f}\left(s^2-\frac{\rho}{G}\right)(\hat{\mathbf{y}}\times\hat{\mathbf{k}})e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SV-TM}, \tag{109}$$

$$\boldsymbol{\tau}=i\omega sG[\hat{\mathbf{k}}(\hat{\mathbf{y}}\times\hat{\mathbf{k}})+(\hat{\mathbf{y}}\times\hat{\mathbf{k}})\hat{\mathbf{k}}]e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SV-TM}, \tag{110}$$

$$p=0, \tag{111}$$

$$\mathbf{E}=i\omega\mu\ \frac{\tilde{\rho}}{\rho_f}LG\beta_T(\hat{\mathbf{y}}\times\hat{\mathbf{k}})e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SV-TM}, \tag{112}$$

$$\mathbf{H}=i\omega s\ \frac{\tilde{\rho}}{\rho_f}LG\beta_T\hat{\mathbf{y}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{SV-TM}, \tag{113}$$

where

$$\hat{\mathbf{k}}(\hat{\mathbf{y}}\times\hat{\mathbf{k}})+(\hat{\mathbf{y}}\times\hat{\mathbf{k}})\hat{\mathbf{k}}=\begin{bmatrix} 2qp/s^2 & 0 & (q^2-p^2)/s^2 \\ 0 & 0 & 0 \\ (q^2-p^2)/s^2 & 0 & -2qp/s^2 \end{bmatrix} \tag{114}$$

and where $s=s_s$ for $SV$ waves and $s=s_{em}$ for $TM$ waves. Note that in both of these transverse-wave polarizations, there are no induced changes in the fluid pressure, but there is induced relative flow due to grain accelerations. This relative flow results in a streaming current that produces the **H** field. Although there is no charge separation produced by the transverse waves, there is a small **E** field generated from the **H** field by induction. For the shear waves, these **E** and **H** fields are carried along with the shear wave as part of the material response and have no support outside the extent of the shear wave. In homogeneous media, shear waves do not generate EM waves. Similarly, for the EM waves, the mechanical components are carried along with the EM waves but have no support outside the extent of the EM wave. In homogeneous media, EM waves do not generate mechanical shear waves.

Finally, the longitudinal $Pf-Ps$ response is

$$\mathbf{u}=\hat{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{Pf-Ps}, \tag{115}$$

$$\mathbf{w}=\beta_L\hat{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{Pf-Ps}, \tag{116}$$

$$\boldsymbol{\tau}=i\omega s[(H-2G+C\beta_L)\mathbf{I}+2G\hat{\mathbf{k}}\hat{\mathbf{k}}]e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{Pf-Ps}, \tag{117}$$

$$-p=i\omega s(C+M\beta_L)e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{Pf-Ps}, \tag{118}$$

$$\mathbf{E}=i\omega(\tilde{\rho}L/\tilde{\epsilon})\beta_L\hat{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{r}}/\mathscr{S}^{Pf-Ps}, \tag{119}$$

$$\mathbf{H}=0, \tag{120}$$

where

$$\beta_L=-\frac{Hs^2-\rho}{Cs^2-\rho_f}, \tag{121}$$

$$\hat{\mathbf{k}}\hat{\mathbf{k}}=\begin{bmatrix} p^2/s^2 & 0 & pq/s^2 \\ 0 & 0 & 0 \\ pq/s^2 & 0 & q^2/s^2 \end{bmatrix} \tag{122}$$

and where $s=s_{pf}$ for fast waves and $s=s_{ps}$ for slow waves. Note that $\mathbf{H}=0$; i.e., there is no net electrical current inside of the compressional waves (the streaming current and conduction current exactly balance) as is easily verified using the other components of the response. The electric field is due entirely to the charge separation associated with the accumulation and depletion of the counterions in the compressions and rarefactions of the waves as already discussed. In homogeneous media, the compressional waves carry this **E** field along with them as part of the material response but do not generate EM waves.

Finally, we conclude with two suggestions for the real scale factors $\mathscr{S}$. It is usually desirable to normalize the above responses so that each plane-wave type carries a unit amount of some specified physical quantity when the wave is at the origin ($\mathbf{r}=0$). Note that because $s$ is complex in general, this quantity will vary when the plane wave is no longer at the origin due to attenuation. A first suggestion is to select $\mathscr{S}$ so that the plane-waves have a unit total-displacement amplitude at the origin; i.e.,

$$|\mathbf{u}+\mathbf{w}|=1 \tag{123}$$

at $\mathbf{r}=0$. The other is to select $\mathscr{S}$ so that the plane waves have a unit of real power flux at the origin; i.e.,

$$\mathrm{Re}\{\hat{\mathbf{k}}\}\cdot\mathrm{Re}\{\mathbf{E}\times\mathbf{H}^*-i\omega(\mathbf{u}^*\cdot\boldsymbol{\tau}-\mathbf{w}^*p)\}=1 \tag{124}$$

at $\mathbf{r}=0$ [recall Eq. (22)]. We do not pause to give the detailed expressions for these scale factors since they are most conveniently determined on the computer [using the responses given above and Eqs. (123) or (124)] in the course of a numerical calculation.

### B. Point-source response

We now consider the solution to Eqs. (74)–(76) when the source terms are given as

$$\mathbf{C}=\mathbf{C}_0\,\delta(\mathbf{r}-\mathbf{r}'), \tag{125}$$

$$\mathbf{F}=\mathbf{F}_0\,\delta(\mathbf{r}-\mathbf{r}'), \tag{126}$$

$$\mathbf{f}=\mathbf{f}_0\,\delta(\mathbf{r}-\mathbf{r}'). \tag{127}$$

The response is represented in terms of Green's tensors as

$$\mathbf{u}=\mathbf{G}_u^F\cdot\mathbf{F}_0+\mathbf{G}_u^f\cdot\mathbf{f}_0+\mathbf{G}_u^C\cdot\mathbf{C}_0, \tag{128}$$

$$\mathbf{w}=\mathbf{G}_w^F\cdot\mathbf{F}_0+\mathbf{G}_w^f\cdot\mathbf{f}_0+\mathbf{G}_w^C\cdot\mathbf{C}_0, \tag{129}$$

$$\mathbf{E}=\mathbf{G}_E^F\cdot\mathbf{F}_0+\mathbf{G}_E^f\cdot\mathbf{f}_0+\mathbf{G}_E^C\cdot\mathbf{C}_0. \tag{130}$$

In isotropic and homogeneous media, the Green's tensors must possess spherical symmetry; i.e., the tensors must depend only on $x=|\mathbf{r}-\mathbf{r}'|$ and they must be symmetric. Thus from our previous symmetry relations [Eqs. (64)–(69)], we have that

$$\mathbf{G}_w^F=\mathbf{G}_u^f, \tag{131}$$

$$\mathbf{G}_E^F=-i\omega\mathbf{G}_u^C, \tag{132}$$

$$\mathbf{G}_E^f=-i\omega\mathbf{G}_w^C, \tag{133}$$

for isotropic and homogeneous media. It is thus only necessary to explicitly solve for the six Green's tensors $\mathbf{G}_u^F$, $\mathbf{G}_w^F$, $\mathbf{G}_E^F$, $\mathbf{G}_w^f$, $\mathbf{G}_E^f$, and $\mathbf{G}_E^C$.

The Green's tensors are found using Fourier transforms where the transform pair is defined

$$\tilde{\mathbf{u}}(\mathbf{k})=\int\mathbf{u}(\mathbf{r})e^{-i\mathbf{k}\cdot\mathbf{r}}d^3\mathbf{r}, \tag{134}$$

$$\mathbf{u}(\mathbf{r})=\frac{1}{(2\pi)^3}\int\tilde{\mathbf{u}}(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}}d^3\mathbf{k}. \tag{135}$$

Upon Fourier transformation, Eqs. (74)–(76) are written in the matrix form

$$\begin{bmatrix} (k^2-\omega^2\mu\tilde{\epsilon})\,\mathbf{I}-\mathbf{kk} & i\omega^3\mu\tilde{\rho}\,L\mathbf{I} & \mathbf{0} \\ -i\omega\tilde{\rho}\,L\mathbf{I} & \omega^2\tilde{\rho}\,\mathbf{I}-M\mathbf{kk} & \omega^2\rho_f\mathbf{I}-C\mathbf{kk} \\ \mathbf{0} & \omega^2\rho_f\mathbf{I}-C\mathbf{kk} & (-Gk^2+\omega^2\rho)\mathbf{I}-(H-G)\mathbf{kk} \end{bmatrix}\begin{bmatrix}\tilde{\mathbf{E}}\\ \tilde{\mathbf{w}}\\ \tilde{\mathbf{u}}\end{bmatrix}=\begin{bmatrix} i\omega\mu\mathbf{C}_0\\ -\mathbf{f}_0\\ -\mathbf{F}_0\end{bmatrix}. \tag{136}$$

To start with, the response $[\tilde{\mathbf{E}}^F, \tilde{\mathbf{w}}^F, \tilde{\mathbf{u}}^F]^T$ associated with $\mathbf{F}_0$ is obtained. Setting $\mathbf{C}_0=0$ and $\mathbf{f}_0=0$, Eq. (136) is put into upper-triangular form and then solved using the identity

$$[\mathbf{I}-\alpha\mathbf{kk}]^{-1}=\mathbf{I}-\frac{\mathbf{kk}}{k^2-1/\alpha}. \tag{137}$$

After a somewhat lengthy (yet straightforward) algebraic manipulation, one finds

$$\tilde{\mathbf{G}}_u^F(\mathbf{k})=\frac{k^2-\omega^2\mu\tilde{\epsilon}(1+\tilde{\rho}\,L^2/\,\tilde{\epsilon})}{G(k^2-\omega^2s_s^2)(k^2-\omega^2s_{em}^2)}\left(\mathbf{I}-\frac{\mathbf{kk}}{k^2}\right)+\left(\frac{M}{MH-C^2}\right)\frac{k^2-\omega^2\tilde{\rho}(1+\tilde{\rho}\,L^2/\,\tilde{\epsilon})/M}{(k^2-\omega^2s_{pf}^2)(k^2-\omega^2s_{ps}^2)}\frac{\mathbf{kk}}{k^2}, \tag{138}$$

$$\tilde{\mathbf{G}}_w^F(\mathbf{k})=-\frac{\rho_f}{\tilde{\rho}}\frac{k^2-\omega^2\mu\tilde{\epsilon}}{G(k^2-\omega^2s_s^2)(k^2-\omega^2s_{em}^2)}\left(\mathbf{I}-\frac{\mathbf{kk}}{k^2}\right)-\left(\frac{C}{MH-C^2}\right)\frac{k^2-\omega^2\rho_f/C}{(k^2-\omega^2s_{pf}^2)(k^2-\omega^2s_{ps}^2)}\frac{\mathbf{kk}}{k^2}, \tag{139}$$

$$\tilde{\mathbf{G}}_E^F(\mathbf{k})=\frac{i\omega^3\mu\rho_fL}{G(k^2-\omega^2s_s^2)(k^2-\omega^2s_{em}^2)}\left(\mathbf{I}-\frac{\mathbf{kk}}{k^2}\right)-i\omega\,\frac{\tilde{\rho}L}{\tilde{\epsilon}}\left(\frac{C}{MH-C^2}\right)\frac{k^2-\omega^2\rho_f/C}{(k^2-\omega^2s_{pf}^2)(k^2-\omega^2s_{ps}^2)}\frac{\mathbf{kk}}{k^2}. \tag{140}$$

To find the response associated with $\mathbf{f}_0$, it is advised to interchange rows two and three of Eq. (136) and then interchange columns two and three so that $[\tilde{\mathbf{E}}^f, \tilde{\mathbf{u}}^f, \tilde{\mathbf{w}}^f]^T$ is the unknown wave field vector. Upon setting $\mathbf{C}_0=0$ and $\mathbf{F}_0=0$, manipulating into upper triangular form and solving, one finds

$$\tilde{\mathbf{G}}_w^f(\mathbf{k})=\frac{(k^2-\omega^2\rho/G)(k^2-\omega^2\mu\tilde{\epsilon}\,)}{\omega^2\tilde{\rho}(k^2-\omega^2s_s^2)(k^2-\omega^2s_{em}^2)}\left(\mathbf{I}-\frac{\mathbf{kk}}{k^2}\right)+\left(\frac{H}{MH-C^2}\right)\frac{k^2-\omega^2\rho/H}{(k^2-\omega^2s_{pf}^2)(k^2-\omega^2s_{ps}^2)}\frac{\mathbf{kk}}{k^2}, \tag{141}$$

$$\widetilde{\mathbf{G}}_E^f(\mathbf{k}) = i\omega\mu L \frac{k^2-\omega^2\rho/G}{(k^2-\omega^2 s_s^2)(k^2-\omega^2 s_{em}^2)} \left(\mathbf{I}-\frac{\mathbf{kk}}{k^2}\right) + i\omega \frac{\widetilde{\rho} L}{\widetilde{\epsilon}} \left(\frac{H}{MH-C^2}\right) \times \frac{k^2-\omega^2\rho/H}{(k^2-\omega^2 s_{pf}^2)(k^2-\omega^2 s_{ps}^2)} \frac{\mathbf{kk}}{k^2}. \tag{142}$$

Lastly, $\widetilde{\mathbf{G}}_E^C$ is obtained as

$$\widetilde{\mathbf{G}}_E^C(\mathbf{k}) = i\omega\mu \frac{k^2-\omega^2\rho_t/G}{(k^2-\omega^2 s_s^2)(k^2-\omega^2 s_{em}^2)} \left(\mathbf{I}-\frac{\mathbf{kk}}{k^2}\right) - i\omega \left(\frac{\widetilde{\rho} L}{\widetilde{\epsilon}}\right)^2 \left(\frac{H}{MH-C^2}\right) \times \frac{k^2-\omega^2\rho/H}{(k^2-\omega^2 s_{pf}^2)(k^2-\omega^2 s_{ps}^2)} \frac{\mathbf{kk}}{k^2}. \tag{143}$$

These are the six independent Green's tensors for a homogeneous and isotropic wholespace.

The spatial response is now determined by taking the inverse-Fourier transform in spherical $\mathbf{k}$ coordinates. The radial $k$ integral is obtained using the residue theorem. We have verified that there is no contribution from the poles at $k=0$ (thus providing a check on the validity of the above expressions). The final spatial response is separated into transverse, longitudinal, and near-field contributions as

$$\mathbf{G}_\eta^\xi(\mathbf{r}|\mathbf{r}') = \sum_{s=s_s}^{s_{em}} T_{\eta,s}^\xi \frac{e^{i\omega s x}}{4\pi x} (\mathbf{I}-\hat{\mathbf{x}}\hat{\mathbf{x}}) + \sum_{s=s_{pf}}^{s_{ps}} L_{\eta,s}^\xi \frac{e^{i\omega s x}}{4\pi x} \hat{\mathbf{x}}\hat{\mathbf{x}} + \left[ \sum_{s=s_s}^{s_{em}} T_{\eta,s}^\xi \left(\frac{i}{\omega s x} - \frac{1}{(\omega s x)^2}\right) \frac{e^{i\omega s x}}{4\pi x} - \sum_{s=s_{pf}}^{s_{ps}} L_{\eta,s}^\xi \left(\frac{i}{\omega s x} - \frac{1}{(\omega s x)^2}\right) \frac{e^{i\omega s x}}{4\pi x}\right] (\mathbf{I}-3\hat{\mathbf{x}}\hat{\mathbf{x}}), \tag{144}$$

where, in this expression,

$$\hat{\mathbf{x}} = \frac{\mathbf{r}-\mathbf{r}'}{x}, \quad x=|\mathbf{r}-\mathbf{r}'| \tag{145}$$

and where $\xi=F$, $f$, or $C$ indicates source type and $\eta=u$, $w$, or $E$ indicates field type. The tensorial polarization $\mathbf{I}-\hat{\mathbf{x}}\hat{\mathbf{x}}$ corresponds to transverse waves while $\hat{\mathbf{x}}\hat{\mathbf{x}}$ corresponds to longitudinal waves. Thus as usual, the near-field polarization $\mathbf{I}-3\hat{\mathbf{x}}\hat{\mathbf{x}}$ can be thought of as a combination of transverse and longitudinal modes. At distances large compared to wavelengths ($\omega s x \gg 1$), the near-field terms become negligible. We now list the various transverse $T_{\eta,s}^\xi$, and longitudinal $L_{\eta,s}^\xi$ amplitudes:

$$T_{u,s_s}^F = \frac{s_s^2-\mu\widetilde{\epsilon}\,(1+\widetilde{\rho}\, L^2/\widetilde{\epsilon}\,)}{G(s_s^2-s_{em}^2)}, \tag{146}$$

$$T_{u,s_{em}}^F = \frac{s_{em}^2-\mu\widetilde{\epsilon}\,(1+\widetilde{\rho}\, L^2/\widetilde{\epsilon}\,)}{G(s_{em}^2-s_s^2)}, \tag{147}$$

$$L_{u,s_{pf}}^F = \left(\frac{M}{MH-C^2}\right) \frac{[s_{pf}^2-\widetilde{\rho}\,(1+\widetilde{\rho}\, L^2/\widetilde{\epsilon}\,)/M]}{(s_{pf}^2-s_{ps}^2)}, \tag{148}$$

$$L_{u,s_{ps}}^F = \left(\frac{M}{MH-C^2}\right) \frac{[s_{ps}^2-\widetilde{\rho}(1+\widetilde{\rho}\, L^2/\widetilde{\epsilon}\,)/M]}{(s_{ps}^2-s_{pf}^2)}, \tag{149}$$

$$T_{w,s_s}^F = -\frac{\rho_f}{\widetilde{\rho}} \frac{(s_s^2-\mu\widetilde{\epsilon}\,)}{G(s_s^2-s_{em}^2)}, \tag{150}$$

$$T_{w,s_{em}}^F = -\frac{\rho_f}{\widetilde{\rho}} \frac{(s_{em}^2-\mu\widetilde{\epsilon}\,)}{G(s_{em}^2-s_s^2)}, \tag{151}$$

$$L_{w,s_{pf}}^F = -\left(\frac{C}{MH-C^2}\right) \frac{(s_{pf}^2-\rho_f/C)}{(s_{pf}^2-s_{ps}^2)}, \tag{152}$$

$$L_{w,s_{ps}}^F = -\left(\frac{C}{MH-C^2}\right) \frac{(s_{ps}^2-\rho_f/C)}{(s_{ps}^2-s_{pf}^2)}, \tag{153}$$

$$T_{E,s_s}^F = \frac{i\omega^3\mu\rho_f L}{G(s_s^2-s_{em}^2)}, \tag{154}$$

$$T_{E,s_{em}}^F = \frac{i\omega^3\mu\rho_f L}{G(s_{em}^2-s_s^2)}, \tag{155}$$

$$L_{E,s_{pf}}^F = -i\omega \frac{\widetilde{\rho}\, L}{\widetilde{\epsilon}} \left(\frac{C}{MH-C^2}\right) \frac{(s_{pf}^2-\rho_f/C)}{(s_{pf}^2-s_{ps}^2)}, \tag{156}$$

$$L_{E,s_{ps}}^F = -i\omega \frac{\widetilde{\rho}\, L}{\widetilde{\epsilon}} \left(\frac{C}{MH-C^2}\right) \frac{(s_{ps}^2-\rho_f/C)}{(s_{ps}^2-s_{pf}^2)}, \tag{157}$$

$$T_{w,s_s}^f = -\frac{(s_s^2-\rho/G)(s_s^2-\mu\widetilde{\epsilon}\,)}{\omega^2\widetilde{\rho}(s_s^2-s_{em}^2)}, \tag{158}$$

$$T_{w,s_{em}}^f = -\frac{(s_{em}^2-\rho/G)(s_{em}^2-\mu\widetilde{\epsilon}\,)}{\omega^2\widetilde{\rho}(s_{em}^2-s_s^2)}, \tag{159}$$

$$L_{w,s_{pf}}^f = \left(\frac{H}{MH-C^2}\right) \frac{(s_{pf}^2-\rho/H)}{(s_{pf}^2-s_{ps}^2)}, \tag{160}$$

$$L_{w,s_{ps}}^f = \left(\frac{H}{MH-C^2}\right) \frac{(s_{ps}^2-\rho/H)}{(s_{ps}^2-s_{pf}^2)}, \tag{161}$$

$$T_{E,s_s}^f = i\omega\mu L \frac{(s_s^2-\rho/G)}{s_s^2-s_{em}^2}, \tag{162}$$

$$T_{E,s_{em}}^f = i\omega\mu L \frac{(s_{em}^2-\rho/G)}{s_{em}^2-s_s^2}, \tag{163}$$

$$L_{E,s_{pf}}^f = i\omega \frac{\widetilde{\rho}\, L}{\widetilde{\epsilon}} \left(\frac{H}{MH-C^2}\right) \frac{(s_{pf}^2-\rho/H)}{(s_{pf}^2-s_{ps}^2)}, \tag{164}$$

$$L_{E,s_{ps}}^f = i\omega \frac{\widetilde{\rho} L}{\widetilde{\epsilon}} \left(\frac{H}{MH-C^2}\right) \frac{(s_{ps}^2-\rho/H)}{(s_{ps}^2-s_{pf}^2)}, \tag{165}$$

$$T_{E,s_s}^C = i\omega\mu \frac{(s_s^2-\rho_t/G)}{s_s^2-s_{em}^2}, \tag{166}$$

$$T_{E,s_{em}}^C = i\omega\mu \frac{(s_{em}^2-\rho_t/G)}{s_{em}^2-s_s^2}, \tag{167}$$

$$L^{C}_{E,s_{pf}} = -i\omega\left(\frac{\widetilde{\rho}\,L}{\widetilde{\epsilon}}\right)^2\left(\frac{H}{MH-C^2}\right)\frac{(s_{pf}^2-\rho/H)}{(s_{pf}^2-s_{ps}^2)}, \tag{168}$$

$$L^{C}_{E,s_{ps}} = -i\omega\left(\frac{\widetilde{\rho}L}{\widetilde{\epsilon}}\right)^2\left(\frac{H}{MH-C^2}\right)\frac{(s_{ps}^2-\rho/H)}{(s_{ps}^2-s_{pf}^2)}. \tag{169}$$

This completes the specification of the Green's tensors.

We have confirmed by direct calculation that Eqs. (131)–(133) are satisfied thus providing another check on the validity of the above expressions. It is seen that from any one of the sources (**F**, **f**, or **C**), all four wave types are generated (shear, EM, fast compressional, and slow compressional). Setting $L=0$ in the above gives the Green's tensors associated with Maxwell's and Biot's equations. Previous work on the point-source response of Biot theory (e.g., Burridge and Vargas, 1979; Norris, 1985) forgot to include the important contributions from **f**.

## III. BOUNDARY CONDITIONS

In this section, the general boundary conditions that must be satisfied at a conformal interface between two dissimilar porous materials are established.

In the section on uniqueness, it was seen that so long as the field components in Eq. (35) are specified on the surface that encloses some region of interest, the wave fields are uniquely determined. However, uniqueness does not help in any way to determine what these specific boundary values should be. The boundary conditions at internal surfaces must come from conservation laws and kinematic constraints as will now be outlined. It will be seen that the boundary conditions so obtained do not violate the uniqueness conditions.

Imagine a porous material 1 conformally overlying another porous material 2. At the interface (denoted by $S_I$), the grains from the two materials come into contact. In linear wave problems, no distinction need be made between welded and nonwelded grain contacts so long as the overlying layer is more than a few tens of centimeters thick. In this case, the static normal force on nonwelded contacts (due to gravity) is significantly greater than the wave-induced deviations, so that the interface grains remain in contact during wave passage.

This leads to the first boundary condition. It is assumed that the grains of material 1 do not penetrate into or slip relative to material 2. This kinematic constraint is expressed (using the porous-continuum variables) as

$$\mathbf{u}_1 = \mathbf{u}_2 \quad \text{on } S_I. \tag{170}$$

The remaining boundary conditions come from Maxwell's equations and the conservation laws governing momentum flux, fluid-mass flux, and energy flux.

In what follows, we use the standard device of integrating Maxwell's equations and the conservations laws over a thin disk of volume $V_D$ that straddles the interface in the limit as the disk thickness vanishes. Upon integrating Eqs. (1) and (2) (the porous-continuum Maxwell's equations) and using Green's theorem, one finds that

$$\mathbf{n}\times[\mathbf{E}_2-\mathbf{E}_1]=0, \tag{171}$$

$$\mathbf{n}\times[\mathbf{H}_2-\mathbf{H}_1]=0 \quad \text{on } S_I, \tag{172}$$

where **n** is the normal to the interface directed from material 1 toward material 2. The horizontal components of **E** and **H** are continuous. An exception to this condition arises if it is assumed that the interface carries a free charge per unit area $\mathscr{Q}$ before the disturbances arrive. Such a surface-charge density could arise electrokinetically if there was a background fluid flow traversing the interface. When a seismic wave shakes such a precharged interface, there is an effective surface current $i\omega\mathbf{n}\times\mathbf{n}\times\mathbf{u}\,\mathscr{Q}$ that must be present on the right-hand side of Eq. (172). Such a discontinuity in the horizontal component of the **H** field represents a possible means by which independently propagating EM waves can be generated from a seismic wave at an interface. However, order of magnitude calculations indicate that such a source will typically be quite small compared to the ''standard'' source involving the mismatch in the seismically induced streaming-current density across an initially uncharged interface.

Next, the conservation of total linear momentum [Eq. (3)] is integrated giving

$$\mathbf{n}\cdot[\boldsymbol{\tau}_2-\boldsymbol{\tau}_1]=0 \quad \text{on } S_I, \tag{173}$$

which states that the total-momentum flux into the interface equals the total-momentum flux out. Although it is possible to write Biot's equations as momentum balances involving the momentum-flux rates through the solid $(1-\phi)\boldsymbol{\tau}_s$ and through the fluid $\phi\boldsymbol{\tau}_f$, it is important to note that neither $\mathbf{n}\cdot[(1-\phi)\boldsymbol{\tau}_s]$ nor $\mathbf{n}\cdot[\phi\boldsymbol{\tau}_f]$ are necessarily continuous across an uncharged interface. Only their sum $\mathbf{n}\cdot\boldsymbol{\tau}$ must be continuous. This is because momentum may be transferred from one phase to another across an interface. A possible exception to Eq. (173) again arises if the interface is assumed to carry a free-charge per unit area $\mathscr{Q}$ before the disturbances arrive. In this case, we must replace the right-hand side by the effective body force per unit area $-\mathscr{Q}\mathbf{E}$.

Next, the conservation of fluid mass is integrated. Dividing the fluid compressibility law [Eq. (7)] by $M$ gives the porous-continuum statement of the conservation of fluid mass

$$\nabla\cdot\mathbf{w} = -\frac{1}{M}\,p - \frac{\mathscr{C}}{M}:\nabla\mathbf{u}. \tag{174}$$

Integrating this over the infinitesimal disk gives

$$\mathbf{n}\cdot[\mathbf{w}_2-\mathbf{w}_1]=0 \quad \text{on } S_I; \tag{175}$$

i.e., fluid flux into the interface equals fluid flux out (recall that $\mathbf{n}\cdot\mathbf{w}$ is the Darcy flux in the direction **n**). There is no way to convert fluid to solid at the interface so that this constraint must always be satisfied. To see that $(\mathscr{C}/M):\nabla\mathbf{u}$ volume integrates to zero at the interface $z=0$ with $\mathbf{n}=\hat{\mathbf{z}}$, we note that in the limit as the disk thickness vanishes

$$\int_{V_D}\left(\frac{\mathscr{C}}{M}:\nabla\mathbf{u}\right)dV$$

$$=\int_{V_D}\left\{\nabla\cdot\left(\frac{\mathscr{C}}{M}\cdot\mathbf{u}\right)-\left[\nabla\cdot\left(\frac{\mathscr{C}}{M}\right)\right]\cdot\mathbf{u}\right\}dV \tag{176}$$

$$=\int_{z=0}\hat{\mathbf{z}}\cdot\left[\left(\frac{\mathscr{C}}{M}\right)_2\cdot\mathbf{u}_2-\left(\frac{\mathscr{C}}{M}\right)_1\cdot\mathbf{u}_1\right]dS$$
$$-\int_{V_D}\delta(z)\hat{\mathbf{z}}\cdot\left[\left(\frac{\mathscr{C}}{M}\right)_2-\left(\frac{\mathscr{C}}{M}\right)_1\right]\cdot\mathbf{u}\;dV \tag{177}$$

$$=0, \tag{178}$$

where $\mathscr{C}/M(\mathbf{r})=(\mathscr{C}/M)_1+[(\mathscr{C}/M)_2-(\mathscr{C}/M)_1]\,H(z)$ with $H(z)$ the step function was used along with continuity of $\mathbf{u}$.

Finally, the conservation of total complex energy [cf. Eqs. (30)–(32)]

$$\nabla\cdot[\mathbf{E}\times\mathbf{H}^*-i\omega(\boldsymbol{\tau}\cdot\mathbf{u}^*-p\mathbf{w}^*)]=Q \tag{179}$$

is integrated over the infinitesimal disk giving that $\mathbf{n}\cdot[\mathbf{E}\times\mathbf{H}^*-i\omega(\boldsymbol{\tau}\cdot\mathbf{u}^*-p\mathbf{w}^*)]$ is continuous across the interface; i.e., the total-energy flux into the interface equals the total-energy flux out. Noting that $\mathbf{n}\cdot(\mathbf{E}\times\mathbf{H}^*)=(\mathbf{n}\times\mathbf{E})\cdot\mathbf{H}^*$, where $\mathbf{n}\times\mathbf{E}$ is both continuous across the interface and parallel to it and that the horizontal components of $\mathbf{H}$ are continuous, one has that $\mathbf{n}\cdot(\mathbf{E}\times\mathbf{H}^*)$ is continuous. Since $\mathbf{n}\cdot\boldsymbol{\tau}$, $\mathbf{u}$, and $\mathbf{n}\cdot\mathbf{w}$ have already been shown to be continuous, one finally arrives at

$$p_1=p_2 \quad \text{on } S_I \tag{180}$$

as the final boundary condition. This is a direct consequence of energy conservation and is always satisfied. If the interface carries an initial surface charge $\mathscr{Q}$, the contribution from the discontinuity in $\mathbf{n}\cdot\boldsymbol{\tau}$ exactly cancels that from $\mathbf{n}\times\mathbf{H}$, and Eq. (180) remains valid. Deresiewicz and Skalak (1963) hypothesize that a possible nonalignment of pores at the interface can result in $p_1$ being different from $p_2$. They argue, essentially, that the resultant mismatch in energy-flux densities is made up for by a dissipation source term due to Darcy flow across the interface. However, the interface, by definition, has no thickness and can produce no dissipation. If the ''interface'' is better characterized as a thin transition zone, then a proper treatment requires the explicit inclusion of a thin porous layer (or ''membrane'') between the two half-spaces. In this case, the boundary conditions at both of the half-space-layer interfaces are as given above. Deresiewicz and Skalak also allow for the possibility that the grains from material 1 exactly plug the pores of material 2 at the interface so that the interface is effectively sealed to Darcy flow. In this case, the boundary condition $p_1=p_2$ is replaced by $\mathbf{n}\cdot\mathbf{w}=0$. However, two granular media can never achieve this situation unless the grains are perfect polyhedra and, even then, only for certain very special packing configurations. This situation never arises in naturally occurring porous materials and need not be worried about in general.

To summarize, at a conformal interface separating two arbitrary porous materials and upon which no free charge is accumulated prior to the arrival of the disturbances, the vector

$$\mathbf{b}=[\mathbf{u},\ \mathbf{n}\cdot\mathbf{w},\quad \mathbf{n}\cdot\boldsymbol{\tau},\quad -p,\ \mathbf{n}\times\mathbf{E},\quad \mathbf{n}\times\mathbf{H}]^T \tag{181}$$

must be continuous. This is the primary result of this section. The continuity of $\mathbf{b}$ satisfies the uniqueness requirements but is in no way a result of uniqueness. At a planar interface, a given incoming plane wave (of any type) will generate, in general, four reflected waves and four transmitted waves (these wave types are shear, EM, fast compressional, and slow compressional). Thus a total of eight boundary conditions are required at a porous–porous interface and the continuity of $\mathbf{b}$ [Eq. (181)] provides exactly these eight equations (note that continuity of the normal and horizontal components of $\mathbf{u}$ and $\mathbf{n}\cdot\boldsymbol{\tau}$ represents four equations).

## IV. CONCLUSIONS

We now briefly summarize a few of the key ideas coming out of this work. In addition to the experimentally measured effect of seismic waves generating electromagnetic waves at interfaces within the Earth, reciprocity has shown that electromagnetic waves from reasonable dipole antennas can generate detectable seismic waves. It may be that under certain conditions it is more convenient to carry out the EM to seismic experiment than the seismic to EM experiment. Reciprocity shows the information content of the two experiments to be equivalent.

Internal to the compressional waves there exist electric fields that act on the diffuse-double-layer charge and can enhance (for slow waves) or retard (for fast waves) the relative flow between the compressions and rarefactions. This effect can alter the attenuation coefficients of the compressional waves by up to 10% for realistic material properties. In shear waves, the electric field is generated by induction and is too small to significantly alter the relative flow. Although the seismic waves generate electromagnetic fields that move along with them as part of the material response, they do not produce independently propagating electromagnetic waves in a homogeneous material. An interface is required for there to be a conversion of wave type.

Because there are three different types of sources in the electroseismic problem ($\mathbf{F}$, $\mathbf{f}$, and $\mathbf{C}$) and three different primary wave fields ($\mathbf{u}$, $\mathbf{w}$, and $\mathbf{E}$) there are nine different Green's tensors required. In homogeneous-isotropic media, these Green's tensors separate into the usual transverse, longitudinal, and near-field polarizations expected in any wave problem. In general, all four wave types (shear, EM, fast compressional, and slow compressional) are generated from each of the three sources ($\mathbf{F}$, $\mathbf{f}$, and $\mathbf{C}$).

Finally, the mechanical boundary conditions at interfaces in porous media are obtained by requiring that: (1) no voids are generated at an interface, (2) the total-momentum flux into an interface equals the total-momentum flux out, (3) the total-fluid-mass flux into an interface equals the total-fluid-mass flux out, and (4) the total-energy flux into the interface equals the total-energy flux out. Details of how pores are aligned at the interface are unimportant.

We conclude by noting that there is no way to properly model the electromagnetic fields generated by seismic waves without using Biot theory to calculate the amount of induced relative flow. In this sense, the electroseismic problem represents one of the most significant applications of Biot theory. Interestingly, the first quantitative work on the theory of seismic waves in porous media (Frenkel, 1944) had as its goal the explanation of electroseismic phenomena.


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